Anesthesia - Previous Year Questions

Marrow Q-Bank Collection • NEET PG / INI-CET / AIIMS (2017-2025)

Total Questions: 110
High-Yield Study Edition

Q1. Anesthesia AIIMS 2019 (May & Nov) ID: MC7742

Which is not an obvious advantage of high pressure oxygen through nasal cannula:

A ✓ Decreases need of intubation

B Reduces nasopharyngeal dead space

C Heating and humidification

D PEEP

Correct Answer: Decreases need of intubation

High flow nasal cannula (HFNC), which delivers oxygen at high flow, and pressure through the nasal cannula.

Advantages of high-flow nasal cannula:

- **Soft and pliable nasal prongs** provide comfort to the patient.
- **Warming and humidification of secretions:** The heating of air to core temperature and humidification of the inspired air is better at high-flow. The humidification causes increased water content in the secretions, which facilitates the removal of secretions. It also decreases the work of breathing and avoids airway dessication and epithelial injury.
- **Reduces dead space:** The high flow of the oxygen delivered clears the upper airway of the expired air, which reduces dead space by a decrease of rebreathing and allows a higher fraction of minute ventilation to participate in alveolar gas exchange.
- **Continuous positive airway pressure (CPAP):** HFNC increases the positive end-expiratory pressure (PEEP), and reduces the work of breathing.
- High flow rate causes **decreased entrainment of room air**, allowing more accurate delivery of oxygen.

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References: Miller's Anesthesia 8th edition, Page no 2883
Online Resource, https://www.uptodate.com/contents/heated-and-humidified-high-flow-nasal-oxygen-in-adults-practical-considerations-and-potential-applications
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6334477/
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5283847/
Miller's Anesthesia 9th edition, Page no. 2543

Q2. Anesthesia AIIMS 2019 (May & Nov) ID: MC7741

Duration of action of Proparacaine:

A 2 min

B 5 min

C 40 min

D ✓ 20 min

Correct Answer: 20 min

The duration of action of proparacaine is approximately **10-20 minutes**.

Proparacaine hydrochloride ophthalmic solution is used as a topical anesthetic agent.

Indications: For procedures which require corneal anesthesia of short duration such as:

- Tonometry
- Gonioscopy
- Removal of corneal foreign bodies
- For short corneal and conjunctival procedures

Contraindication: Patients with known hypersensitivity

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References: Online Resource, https://dailymed.nlm.nih.gov/dailymed/fda/fdaDrugXsl.cfm?setid=447a2a64-0507-4fe8-bb44-cc36c3916151&type=display
Online Resource, https://reference.medscape.com/drug/alcaine-ophthaine-proparacaine-ophthalmic-343639#10

Q3. Anesthesia AIIMS 2019 (May & Nov) ID: MD0243

Glycopyrrolate is used as a pre-anesthetic agent for:

A ✓ Reducing the secretions

B Sedative effect

C Skeletal muscle relaxant

D Anxiolytic

Correct Answer: Reducing the secretions

Glycopyrrolate is used as a pre-anesthetic agent for **reducing the secretions.**

Anticholinergic drugs e.g. atropine, glycopyrrolate are routinely administered with anticholinesterases in order **to attenuate the undesirable muscarinic effects.** Glycopyrrolate is usually preferred because it has less vagolytic effects than atropine at doses that inhibit secretions and does not cross the blood-brain barrier. It should be administered as early as possible to maximize its effectiveness.

Glycopyrrolate 0.2 mg IV, is administered to dry airway secretions before flexible scope intubation of the trachea, during the bronchoscopic examination.

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References: Miller's Anesthesia 8th edition, Page nos 1012, 1657, 1672, 1978

Q4. Anesthesia AIIMS 2019 (May & Nov) ID: MD0244

Identify the device shown in the image below.



A Automated External Desynchonizer

B ✓ Automated External Defibrillator

C Automated Electronic Defibrillator

D Automated External decompressor

Correct Answer: Automated External Defibrillator

The device is an **Automated External Defibrillator (AED).**

This is a special medical device that is commonly used for recognizing the absence/presence of **rapid ventricular tachycardia or ventricular fibrillation.** With no need for intervention by an operator, it determines whether an electrical shock right to the heart must be performed (defibrillation).

An automated external defibrillator can be used on those victims, who are unconscious and aren't breathing in a normal way. As a rule, it is used together with CPR in cases of sudden cardiac arrest. It is forbidden to use it on patients who are less than eight years old or those weighing less than 25 kgs.

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References: Online Resource, https://medinstrum.com/automated-external-defibrillator/
Miller's Anesthesia 8th edition, P 3186, 2877

Q5. Anesthesia AIIMS 2019 (May & Nov) ID: MD0245

You are an intern posted in the ICU. While taking a blood sample for ABG, which of the following steps would you not do?

A Normal ph can't rule out acid base disorder

B ✓ Add 0.5 ml Heparin to the syringe before taking blood

C If modified Allens test is negative, alternate site is chosen

D Radial artery is the preferred site

Correct Answer: Add 0.5 ml Heparin to the syringe before taking blood

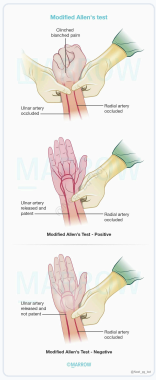
0.5 ml of Heparin is not added to the syringe before taking blood.

Very little **heparin** is needed in the sample to prevent clotting; **0.05 to 0.10 ml of a dilute solution** will help prevent coagulation of 1 ml of blood without affecting its pH. This is achieved by drawing 0.5 ml into the syringe and **flushing it out.** The residue that remains adherent to the syringe is sufficient for the procedure.

Heparin must be added to the syringe as an **anticoagulant**. Because the pH of heparin is near 7.0, and the Po_2 and Pco_2 of the heparin solution are near room air values, excess heparin can alter ABG measurements. After flushing the syringe with heparin, a sufficient amount usually remains in the dead space of the syringe and needle for anticoagulation without distortion of the ABG determination.

Note:

- Modified Allen's test positive indicates that the collateral blood flow is normal
- Modified Allen's test negative indicates that the collateral blood flow is abnormal and an alternate site must be chosen.



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References: Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK371/
Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK138652/
Online Resource, https://www.euro.who.int/__data/assets/pdf_file/0005/268790/WHO-guidelines-on-drawing-blood-best-practices-in-phlebotomy-Eng.pdf
Online Resource, https://www.annalsthoracicsurgery.org/article/S0003-4975(99)00095-8/pdf

Q6. Anesthesia AIIMS 2019 (May & Nov) ID: MD0246

Which of the following is not a risk factor for postoperative pulmonary complications?

A BMI > 30

B ✓ Well controlled asthma

C Upper abdominal surgery

D Age > 70 years

Correct Answer: Well controlled asthma

Well-controlled asthma is not a risk factor for postoperative pulmonary complications.

Risk factors for postoperative pulmonary complications:

1. Age >70 years
2. COPD (uncontrolled)
3. Neck, upper abdominal, Aortic, neurologic surgical procedures
4. Prolonged procedure >2 hours
5. General anesthesia with intubation
6. BMI > 30
7. Albumin concentration < 35 g/L
8. Inability to walk two blocks on level land or one flight of stairs

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References: Miller's Anesthesia 8th edition, Page No. 1114
Miller's Anesthesia 9th edition, Page No. 954

Q7. Anesthesia AIIMS 2019 (May & Nov) ID: MD7334

Which of the following is associated with respiratory alkalosis?

- A ✓ Assisted control mode ventilation
- B Controlled mandatory
- C Pressure controlled
- D SIMV

Correct Answer: Assisted control mode ventilation

Respiratory alkalosis can occur in Assist Control Mode Ventilation

Ventilatory Mode	ACMV (Assist Control Mode Ventilation)	IMV (Intermittent Mandatory Ventilation)	PSV (Pressure Support Ventilation)	NIV (Noninvasive Ventilation)
Variables set by User (Independent)	Tidal volume Ventilator rate Fio2 PEEP level Pressure limit	Tidal volume Mandatory ventilator rate Fio2 PEEP level Pressure limit Spontaneous breaths between assisted breaths	Inspiratory pressure level Fio2 PEEP Pressure limit	Inspiratory and expiratory pressure level Fio2
Variables Monitored by the user (Dependent)	Peak, mean, and plateau airway pressures VE ABG I/E ratio	Peak, mean, and plateau airway pressures VE ABG I/E ratio	Tidal volume Respiratory rate VE ABG	Tidal volume Respiratory rate VE ABG
Trigger cycle limit	Patient effort Timer Pressure limit	Patient effort Timer Pressure limit	Pressure limit Inspiratory flow	Pressure limit Inspiratory flow
Advantages	Patient control Guaranteed ventilation	Patient control Comfort from spontaneous breaths Guaranteed ventilation	Patient control Comfort Assures synchrony	Patient control
Disadvantages	Potential hyperventilation **(Leading to respiratory alkalosis)** Barotrauma and volutrauma Every effective breath generates a ventilator volume	Potential dysynchrony Potential hypoventilation	No timer backup Potential hypoventilation	Mask interface may cause discomfort and facial bruising Leaks are common Hypoventilation

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References: USMLE Step 1 Lecture Notes 2018: 7-Book Set (Kaplan Test Prep), Page no 288

Q8. Anesthesia AIIMS 2020 (May and Nov INI-CET) ID: MD4631

A patient of blunt chest trauma has been started on mechanical ventilation. Which of the following is not correct about the initial setting?

A Respiratory rate - 12 Breaths/ min

B PEEP - 5cm H20

C FiO2- 100%

D ✓ Tidal volume - 12ml/kg of actual BW

Correct Answer: Tidal volume - 12ml/kg of actual BW

The **tidal volume of 12ml/kg of actual body weight** is an **incorrect initial setting** for mechanical ventilation.

Initial ventilatory settings are as given below:

- Tidal volume - 4-6 ml/kg of IBW (Ideal Body Weight)
- Respiratory rate - 12 Breaths/ min
- FiO2 - 100% to begin with and titrated to lesser values to prevent oxygen toxicity
- PEEP - Begin with 5 cm H2O, titrate to 10 cm H2O (Carefully increased as high PEEP may impair ventricular filling in a patient with blunt chest trauma)

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References: Tintinalli's Emergency Medicine , 8th Ed, Page no 191
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3132366/
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3132366/

Q9. Anesthesia AIIMS 2020 (May and Nov INI-CET) ID: MD0243

Glycopyrrolate is used as a pre-anesthetic agent for:

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B Sedative effect

C Skeletal muscle relaxant

D Anxiolytic

Correct Answer: Reducing the secretions

Glycopyrrolate is used as a pre-anesthetic agent for **reducing the secretions.**

Anticholinergic drugs e.g. atropine, glycopyrrolate are routinely administered with anticholinesterases in order **to attenuate the undesirable muscarinic effects.** Glycopyrrolate is usually preferred because it has less vagolytic effects than atropine at doses that inhibit secretions and does not cross the blood-brain barrier. It should be administered as early as possible to maximize its effectiveness.

Glycopyrrolate 0.2 mg IV, is administered to dry airway secretions before flexible scope intubation of the trachea, during the bronchoscopic examination.

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References: Miller's Anesthesia 8th edition, Page nos 1012, 1657, 1672, 1978

Q10. Anesthesia AIIMS 2020 (May and Nov INI-CET) ID: MA2738

Cardiac output can be determined using the following except?

A Thermal dilution technique

B ✓ Capnography

C Esophageal doppler

D Echocardiography

Correct Answer: Capnography

Capnography is a measurement of **end-tidal CO_2,** which is used to confirm tracheal intubation and assess ventilation in the perioperative period.

Cardiac output measurements technique:

1. Thermodilution
2. Dye dilution, using indocyanine green
3. Lithium dilution
4. Pulse contour devices
5. Esophageal doppler
6. Thoracic bioimpedance
7. Echocardiography

Note: Thermal dilution method uses a non-toxic liquid of temperature different from the blood. This fluid is injected into a vein and is detected on the arterial side using a catheter with a temperature sensor. The rate of flow of this liquid is used to calculate the cardiac output.

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References: Miller's Anesthesia 8th edition, Page nos 1387-90
Online Resource, https://pubmed.ncbi.nlm.nih.gov/20185659/
Online Resource, https://www.sciencedirect.com/topics/medicine-and-dentistry/thermodilution

Q11. Anesthesia AIIMS 2020 (May and Nov INI-CET) ID: MD6697

Identify the equipment shown in the image below.



A Mechanical ventilator

B Infusion pump

C CPAP machine

D ✓ High flow nasal cannula

Correct Answer: High flow nasal cannula

The equipment shown in the image is high flow nasal cannula.

High flow nasal cannula is a mode of respiratory support used in the management of acute respiratory failure. It delivers **humidified gas** via the nasal cannula at high flow rates.

- HFNC can deliver oxygen at 40L/min, 37°C, and 99.9% relative humidity.
- The **high-flow oxygen** is delivered directly to the nasopharynx. This produces **FiO_2** equal to that delivered by traditional mask devices.
- The benefits of HFNC include **reduced work** of breathing, **improved gas exchange** by nasopharyngeal dead-space washout, and **decreased energy** required to **humidify** and **heat** respiratory gases.

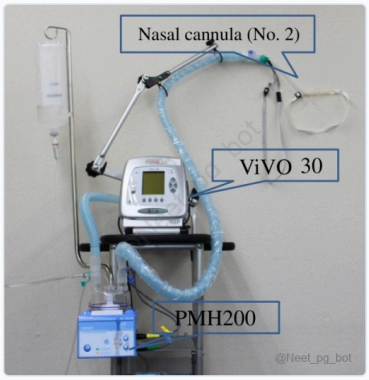
The HFNC equipment shown in the image below uses a heated humidifier (named PMH200), a nasal cannula (no. 2), and a ventilator (brand: vivo30).

Other options:

Option a: Mechanical ventilator

Option b: Infusion Pump

Option c: CPAP









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References: Miller's Anesthesia 9th edition, Page no. 2595, 2752

Q12. Anesthesia AIIMS 2020 (May and Nov INI-CET) ID: MD6698

The blue pipeline in the operation theatre is used for delivering which of the following gases?

A Oxygen

B ✓ Nitrous oxide

C Air

D Vacuum

Correct Answer: Nitrous oxide

The **blue pipeline** in operation theatre is for **nitrous oxide.**

Other options:

Oxygen: White

Air: Black

Vacuum/Central suction: Yellow

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References: Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3821266/
Online Resource, https://www.bochealthcare.co.uk/en/images/cylinder_data_med309965_2011_tcm409-54065.pdf.

Q13. Anesthesia AIIMS 2020 (May and Nov INI-CET) ID: MD6699

The given set of instruments are used in:



A Central line insertion

B Arterial line insertion

C ✓ Airway management

D Umbilical catheter insertion

Correct Answer: Airway management

The image shows a tray in an **airway cart** with the instruments necessary for **airway management.**

An airway cart may include:

- **Laryngoscope blades** of various sizes, including a rigid fibreoptic.
- **Tracheal tubes** of various sizes, and tracheal tube guides such as semirigid stylets, ventilating tube changer, light wands, and forceps that are required to manipulate the distal portion of the tracheal tube.
- **Masks** of various sizes.
- **Laryngeal mask airways** of various sizes.
- Flexible fiberoptic **intubation** equipment.
- Retrograde intubation equipment.
- At least one device to perform emergency **noninvasive airway ventilation** such as esophageal tracheal tube, a hollow jet ventilation stylet.
- Equipment required for emergency **invasive** airway access.
- An exhaled **CO_2 detector.**

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References: Online Resource, https://accessanesthesiology.mhmedical.com/content.aspx?sectionid=41840249&bookid=414

Q14. Anesthesia AIIMS 2020 (May and Nov INI-CET) ID: MD6700

On your morning walk in the park, you notice an unconscious man lying on the ground. Which of the following options would be the next best step?

A Check carotid pulse, start CPR, and call for help; Defibrillate if no pulse.

B ✓ Call for help, check pulse, do CPR, defibrillate if no pulse

C Observe

D Ignore and continue to walk

Correct Answer: Call for help, check pulse, do CPR, defibrillate if no pulse

The next best step is to Call for help, check pulse, do CPR, defibrillate If no pulse.

Steps in basic life support:

First step – Assess safety of the surroundings.

Second step - Assess for responsiveness

Third step – Proceed if the patient is unresponsive. Shout for help.

If the patient is unresponsive, assess for breathing and carotid pulse simultaneously for no more than 10 seconds.

In the given scenario, it is already mentioned that you are in a park (scene safety has been assessed), and that the person is unconscious (responsiveness has been assessed). So, the immediate next step is to call for help, and start CPR.

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References: Online Resource, https://www.acls-pals-bls.com/algorithms/bls/

Q15. Anesthesia AIIMS 2020 (May and Nov INI-CET) ID: MD6916

Which of the following can be considered as effective CPR?

1. Respiratory rate: 20 - 24 breaths per minute.

2. Chest compressions: 100-120/min.

3. Allow complete chest recoil.

4. 5-6 cm depression.

A 1, 2, 3, and 4

B 1, 2, and 4

C ✓ 2, 3, and 4

D 1 and 2 only

Correct Answer: 2, 3, and 4

The ideal rescue breath rate for adults is 10 breaths/min, while in infants and children up to puberty it is 20-30 breaths/min. As the person's age group is not specified in the question, assuming an adult, statement 1 is incorrect, while 2, 3, and 4 are correct.

Effective chest compression in **adults**:

- Provide **100-120 compressions per minute** (30 compressions for every 15-18 sec).
- Place your arms, one over the other in the midline, on the lower 1/3rd of the patient's sternum, and lock your arms.
- Use both arms to press the chest to a depth of **5-6 cm** (2 - 2.4 inches) or more.
- Press hard and fast
- Allow for **full chest recoil** after every compression.

Effective chest compressions in a **child older than 1 year and up to puberty:**

- The compression rate is 100 - 120/min (30 compressions for every 15-18 sec).
- **One or both palms** are placed in the midline.
- The depth of compression is **2 inches.**
- Press hard and fast
- Allow for full chest recoil after every compression.

Effective chest compressions in **infants (<1 year):**

- 100 to 120 compressions per minute (30 compressions for every 15-18 sec).
- **One provider: Place two fingers**, one between the nipple line and the other 1 cm below.
- **Two providers: Encircle** the infant's **torso** with both hands with **both thumbs** pointing cephalic positioned 1cm below the nipples over the sternum.
- Depth of compressions should be at least **1.5 inches or 1/3 the depth** of the infant's chest.
- Press hard and fast
- Allow for full chest recoil after every compression.

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References: Online Resource, https://www.acls-pals-bls.com/algorithms/bls/

Q16. Anesthesia INI-CET 2023 ID: MA2848

A patient who is under general anaesthesia shows the following finding on the ET CO2 monitor. Which condition can lead to this characteristic waveform on capnography?



A Malignant hyperthermia

B ✓ Bronchospasm

C Rebreathing

D Accidental extubation

Correct Answer: Bronchospasm

The above capnographic waveform depicts **bronchospasm**.

Shark fin appearance - Prolonged phase II, increased α angle, and steeper phase III suggest bronchospasm or airway obstruction.

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References: Miller's Anesthesia 8th edition, Page no. 1554
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no. 127, 128
Miller's Anesthesia 9th edition, Page no. 1311

Q17. Anesthesia INI-CET 2023 ID: MF9191

In a patient with thrombocytopenia, platelet transfusion is done when the platelet level drops below a threshold. To proceed with surgery, platelet transfusion is done to achieve a platelet level of?

A 30,000

B 40,000

C ✓ 50,000

D 60,000

Correct Answer: 50,000

In an otherwise healthy patient, **surgery can safely be performed** in patients with **platelet counts higher than 50,000/mm^3**. The risk of bleeding increases significantly if the platelet count is below this level. When platelet transfusions are used to treat thrombocytopenia, the platelet count **rises by 10,000/mm^3 for every unit** transfused.

Thrombocytopenia is defined as a platelet count less than 1,50,000/mm^3. It may be the result of immune thrombocytopenic purpura (ITP), drug-induced thrombocytopenia, rheumatological diseases (SLE, rheumatoid arthritis), pregnancy and chronic liver disease. The initial step if the patient has low platelet count is to repeat the test and examine the peripheral smear.

Anemia is also a common preoperative hematological disorder. The optimal preoperative hemoglobin concentration varies between 7.5 to 10 g/dL across the individuals, with inter-individual differences due to co-morbid conditions in the patients. Patients with anemia or suspected anemia must undergo a battery of investigations to determine the type of anemia and how best to correct it. Preexisting anemia is a risk factor for postoperative death and complications such as AKI, stroke and infections.

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References: Miller's Anesthesia 9th edition, Page 963, 964, 967, 968

Q18. Anesthesia INI-CET 2023 ID: MF9185

The presence of which of the following indicates that CPR is of high quality?

1. **Chest compression at a rate of 100-120/minute**
2. **Breathing rate of 20-25/minute**
3. **Depth of compression should be 5-6 cm**
4. **Allow chest recoil**

A 2, 3 only

B ✓ **1, 3, 4 only**

C 1, 2, 3, 4

D 1 only

Correct Answer: 1, 3, 4 only

Chest compressions at a rate of **100-120/minute**, **depth of compression 5-6 cm**, and allowing **complete chest recoil** indicate **high-quality CPR**. The **compression-to-ventilation ratio** in the absence of an advanced airway should be **30:2** (30 compressions to be followed by 2 breaths). If an **advanced airway** (endotracheal tube or supraglottic airway device) is **present**, **10 breaths/minute** should be given.

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References: Basic Life Support Provider Manual, Page 10
Miller's Anesthesia 9th edition

Q19. Anesthesia INI-CET 2023 ID: MF9179

A 70 kg patient who presented to the emergency with acute pancreatitis was shifted to the ICU and intubated. His ventilatory settings are: FiO2 90%, PEEP 12 mmHg, respiratory rate 16/min, and tidal volume 420 ml. The results of the ABG analysis are given below. Which of the following would be the next step in the management of this patient?

pH: 7.33

PaO_2: 110 mmHg

$PaCO_2$: 58 mmHg

SpO_2: 100%

A Increase PEEP to 16

B Increase tidal volume

C Reduce respiratory rate

D ✓ Reduce FiO2

Correct Answer: Reduce FiO2

In a patient with a history of **acute pancreatitis** who needed to be intubated, the diagnosis of **acute respiratory distress syndrome (ARDS)** is the most likely. In this patient, the **PaO_2** is **110 mmHg** and the **SpO_2** is **100%**, which are significantly **higher than the target** recommended in patients with ARDS.

The next step in the management of this patient would be to **reduce the FiO_2** to **prevent any further lung injury** and worsening of ARDS due to **oxygen toxicity**. In the initial phase of the management, the **FiO_2** of the patient should be kept **≤ 60%** and SpO_2 should be maintained between **88-95%**.

The main treatment strategy which has been shown to reduce mortality in patients is **low tidal volume ventilation** (**6 mL/kg** of predicted body weight). **Increasing the tidal volume** (option B) in this patient will lead to **barotrauma** to the alveoli because air is pushed through the alveoli which are already collapsed.

Although a high PEEP keeps the alveoli opened up, there is **no rationale for increasing the PEEP further** (option A) in this patient because the **oxygenation is adequate**. Furthermore, **PEEP** must be **adjusted to minimize FiO_2** and provide adequate PaO_2 at the same time.

Hypercapnia is **expected** in a patient on **low tidal volume ventilation**. It can be managed by **increasing the respiratory rate**. However, respiratory acidosis is tolerated by the patient up to a blood pH of 7.2. This is known as **permissive hypercapnia**. **Reducing the respiratory rate** (option C) in this patient will increase the $PaCO_2$ further and cause a **worsening of respiratory acidosis**.

Apart from these, **prone position**, **early neuromuscular blockade** (to prevent the patient from fighting the ventilator), and use of **extracorporeal membrane oxygenation (ECMO)** have also been shown to improve mortality in patients with ARDS. Management strategies that have **not been shown to improve the outcome** in ARDS patients include **glucocorticoids**, **surfactant replacement**, anti-inflammatory therapies like NSAIDs, and inhaled vasodilators.

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References: Harrison's Principles of Internal Medicine, 21st Edition, Page 2227-2229, 2233

Q20. Anesthesia INI-CET 2023 ID: MG1070

Which of the following best represents the American Society of Anesthesiologists (ASA) physical status III?

A Mild systemic disease

B Moderate systemic disease

C ✓ Severe systemic disease

D Severe systemic disease with threat to life

Correct Answer: Severe systemic disease

Severe systemic disease best fits into **grade 3** of the ASA classification.

American Society of Anesthesiologists classification is used to define **relative risk** prior to **conscious sedation** and **surgical anesthesia.**

It is simple, reproducible and **strongly correlates to perioperative risk.**

American Society of Anesthesiologists physical status classification of patients-

Class 1: Normal healthy patient
Class 2: Patient with mild systemic disease (**no** functional limitations)
Class 3: Patient with severe systemic disease (some functional limitations)
Class 4: Patient with severe systemic disease that is a constant threat to life (functionality incapacitated)
Class 5: A moribund patient who is not expected to survive without the operation
Class 6: A brain-dead patient whose organs are being removed for donor purposes

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References: Miller's Anesthesia 8th edition, Page no: 1144
Miller's Anesthesia 9th edition, Page no: 987

Q21. Anesthesia INI-CET 2023 ID: MG1095

A 5-year-old is on 3L/min O2 via a nasal cannula. What is the appropriate FiO2 level for this child?

A 25

B ✓ 30

C 45

D 60

Correct Answer: 30

The appropriate FiO2 for this child would be **30.**

The formula for calculating FiO2 for a child on oxygen supplementation through nasal cannula is **FiO2= 21%+ (nasal cannula flow in L/min) × 4**

A nasal cannula is a low flow device, and FiO2 should be adjusted according to flow rates.

Examples of low-flow devices are:

- Nasal cannula
- Face mask
- Face tent
- Non-rebreather mask.

High-flow devices include:

- Venturi mask
- High flow nasal cannula
- Mechanical ventilation

Oxygen delivery devices	Flow rate (liter/min)	Approximate FiO2
Nasal cannula	1-6	0.24-0.44
Simple face mask	6-10	0.40-0.60
Non-rebreathing mask	10-15	0.90-1.00
Venturi mask	2-15	0.24-0.60

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References: Morgan and Mikhail's Clinical Anesthesiology, 6th edition, page 1283
Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK560867/

Q22. Anesthesia INI-CET 2025 ID: MG9312

1. Identify incorrect statements 1. Alcohol at low doses causes CNS stimulation and at high doses causes CNS depression. 2. Cannabis causes repetitive movements (driven) 3. Opioids are effective analgesics 4. Volatile inhalants are toxic

A 3, 4

B ✓ 1, 2

C 2, 3

D 1, 3

Correct Answer: 1, 2

Statements 1 and 2 are incorrect. Alcohol is a CNS depressant at all doses. Although alcohol may appear to cause stimulation at low doses, this effect is due to depression of inhibitory control centres in the brain, not true CNS stimulation. (Statement 1) Cannabis does not produce repetitive movements, which are more characteristic of central nervous system stimulants like amphetamines or cocaine (Statement 2). Cannabis, on the contrary, can impair motor skills. Opioids like morphine, codeine, and fentanyl are highly effective analgesics (Statement 3). Opioids act as analgesics primarily by activating μ-opioid receptors, which inhibit GABA release and enhance PAG output, modulating pain pathways in the brain and spinal cord. At the spinal level, they block calcium channels presynaptically, reducing neurotransmitter release, and open potassium channels postsynaptically, causing hyperpolarization. This dual action dampens the excitation of second-order neurons, effectively reducing pain transmission. Volatile inhalants, such as those in adhesives, butane refills, and aerosol propellants, are highly toxic and can cause sudden death due to cardiac arrhythmias or respiratory depression. Acute abuse leads to rapid intoxication marked by euphoria, hallucinations, and potential cardiac arrest. Chronic use, especially of substances like toluene or n-hexane, can result in lasting damage to the brain, heart, liver, kidneys, and lungs. (Statement 4)

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References: Goodman & Gillman's The Pharmacological Basis Of Therapeutics - 14th Edition • Page 522, 449-451, 232
A Short Textbook of Psychiatry by Niraj Ahuja, 7th Edition • Page 35-42
Online Resource • https://www.researchgate.net/publication/45709888_Substance_Abuse_and_Movement_Disorders
Online Resource • https://pubmed.ncbi.nlm.nih.gov/2222869/
Online Resource • https://pmc.ncbi.nlm.nih.gov/articles/PMC6478951/

Q23. Anesthesia INI-CET 2025 ID: MG9300

2. A patient undergoes cardiac surgery with muscle relaxant Vecuronium. After the surgery is completed, which of the following agents can be used for the reversal of this drug?

A Baclofen

B Dantrolene

C ✓ Neostigmine

D Atropine

Correct Answer: Neostigmine

Neostigmine can be used in the reversal of muscle relaxants like Vecuronium. Neostigmine is an acetylcholine esterase inhibitor. Its primary mechanism of action is decreased breakdown of acetylcholine at the motor endplate. It increases the concentration of acetylcholine in the synaptic cleft, which can compete with and displace the non-depolarising muscle relaxants, and thus reverse the paralysis. Acute bolus or prolonged administration of anticholinesterases can have deleterious effects on neuromuscular function in otherwise healthy patients. Sugammadex is a novel reversal drug that shows a high affinity for the steroidal neuromuscular blocking drugs rocuronium and vecuronium. It is a cyclodextrin that forms a tight inclusion complex with these steroidal neuromuscular blocking drugs, thereby inactivating their effects and facilitating rapid reversal of neuromuscular block. Other options: Option A: Baclofen is a central-acting GABA-B agonist used to treat spasticity. It does not affect neuromuscular junctions or reverse non-depolarising muscle relaxants. Option B: Dantrolene acts directly on skeletal muscle to reduce calcium release and is used for malignant hyperthermia, not for reversing neuromuscular blockade. Option D: Atropine is an antimuscarinic agent used to prevent bradycardia during reversal, but does not reverse neuromuscular blockade by itself. Pearl PM0215: Types of muscle relaxants There are 2 types of muscle relaxants: depolarizing and non-depolarizing. Depolarizing Muscle Relaxant- Succinylcholine Non-Depolarizing Muscle Relaxants: * Ultra short-acting (<10 minutes) - Gantacurium * Short-acting (10-20 minutes) - Mivacurium * Intermediate-acting (20-50 minutes) * Atracurium * Cisatracurium * Vecuronium * Rocuronium * Long-acting (>50 minutes) * Pancuronium * Doxacurium * Pipecuronium

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References: Miller's Anesthesia - Edition 10 • Page no. 1145, 3221

Q24. Anesthesia INI-CET 2025 ID: MG9299

3. Which of the following are true regarding benzodiazepines and barbiturates: 1. Both act on GABA-A chloride channel 2. Alcohol causes superadded CNS depression 3. Flumazenil reverses the action of both 4. Thiopentone has short duration of action due to its rapid metabolism

- A ✓ Both 1 and 2 are correct
- B Only 1 is correct
- C Only 3 is correct
- D All are correct

Correct Answer: Both 1 and 2 are correct

Both statements 1 and 2 are correct regarding benzodiazepines and barbiturates Both benzodiazepines and barbiturates act on the GABA-A chloride channel, but at different sites (Statement 1). Benzodiazepines facilitate the action of GABA by increasing the frequency of opening of the GABA-mediated chloride channel. Examples include alprazolam, clonazepam, lorazepam, midazolam, etc. On the other hand, barbiturates facilitate the action of GABA by increasing the opening duration of the GABA-mediated chloride channel. Examples include amobarbital, phenobarbital, butabarbital, and pentobarbital. Alcohol can cause superadded CNS depression when combined with either drug, increasing the risk of respiratory depression (Statement 2). Flumazenil is a competitive antagonist at the benzodiazepine binding site of the GABA-A receptor. Flumazenil reverses the effects of benzodiazepines only, not barbiturates (Statement 3). No specific antidote exists for barbiturates, and overdose treatment includes supportive care and urinary alkalinization. Thiopentone has a rapid onset of action because of its lipophilic nature, so it rapidly crosses the blood-brain barrier and enters highly perfused organs like the brain, heart, liver, and kidneys. As thiopentone redistributes from the brain to tissues, it perfuses to muscle, fat, and skin. Over time, due to its high lipid solubility, thiopentone accumulates maximally in fat. Fat acts as a storage reservoir since it has low blood flow, and drug release from fat occurs slowly. This redistribution reduces the concentration of thiopentone, terminating its anaesthetic effect. Hence, thiopentone has a short duration of action due to its redistribution, not rapid metabolism (Statement 4).

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References: Goodman & Gillman's The Pharmacological Basis Of Therapeutics - 14th Edition • Page 30, 394, 434, 894, 526, 437
Morgan and Mikhail's Clinical Anesthesiology, 7th edition • Page 322
Online Resource • https://www.ncbi.nlm.nih.gov/sites/books/NBK539731
Online Resource • https://pubmed.ncbi.nlm.nih.gov/29095046

Q25. Anesthesia NEET 2024 ID: MG6761

A patient undergoing surgery in the prone position is maintained on mechanical ventilation with PEEP. What is the significance of using PEEP in this scenario?

- A ✓ Prevents alveoli collapse and atelectasis of lung
- B Decreases inspiratory pressure
- C Increases venous return and maintains cardiac output
- D Reduces CO2 in alveoli

Correct Answer: Prevents alveoli collapse and atelectasis of lung

The significance of using **positive end-expiratory pressure (PEEP)** in the given clinical scenario is to help keep the alveoli **open,** preventing collapse **(atelectasis),** which is especially important in the prone position where certain lung regions might be more susceptible to collapse.

This is crucial in conditions where the lungs are prone to collapse, such as **Acute Respiratory Distress Syndrome (ARDS)** or when the patient is in a **prone position** during surgery. By maintaining open alveoli, PEEP improves gas exchange, which increases oxygen levels in the blood.

Other options:

Option B: Decreases inspiratory pressure: PEEP can **increase** the overall inspiratory pressure by maintaining pressure in the lungs at the end of expiration.

Option C: Increases venous return and maintains cardiac output: PEEP can **decrease** venous return due to increased intrathoracic pressure, which may reduce cardiac output.

Option D: Reduces CO2 in alveoli: PEEP does not directly reduce CO2 in the alveoli. The effect on CO2 depends on ventilation and perfusion matching, not just PEEP alone.

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References: Harrison's Principles of Internal Medicine, 21st Edition, Page no : 2033

Q26. Anesthesia NEET 2024 ID: MG6769

Which of the following statements are true about eutectic mixture of local anesthetic (EMLA)?



A 1gm of 0.5% of cream contains 50mg lignocaine

B ✓ Methohemoglobinemia is a side effect

C It is an equal mixture

D It is made of Lignocaine and ropivacaine

Correct Answer: Methohemoglobinemia is a side effect

Methemoglobinemia can develop when **high doses of prilocaine** are administered, typically over 600 mg.

EMLA (Eutectic Mixture of Local Anaesthetic) is a topical mixture containing prilocaine. The risk of methemoglobinemia is increased in neonates with metabolic disorders or in those receiving drugs that can impair the reduction of methemoglobin.

It is an **eutectic(in fixed proportions)** mixture, not necessarily equal (Option C). It consists of a 1:1 mixture of 2.5% lidocaine (Option A) and 2.5% prilocaine, not ropivacaine (Option D)

EMLA can provide **adequate dermal analgesia** for procedures like:

- Split skin-graft harvesting
- Venipuncture
- Intravenous cannulation in children
- Laser skin treatments
- Lithotripsy
- Circumcision

EMLA cream **should not** be **used** on **mucous membranes, broken skin,** infants younger than 1 month of age, or patients with a predisposition to methemoglobinemia. EMLA is applied 45-60 minutes before the procedure with an occlusive dressing. It provides analgesia up to a depth of 5 mm, which lasts for 1-2 hours.

Other side effects include- **blanching** of the skin, **erythema,** and **oedema.**

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References: Miller's Anesthesia 8th edition, Page nos 1044, 1045 and 1050
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no 270

Q27. Anesthesia NEET 2024 ID: MG6768

An anesthetist used the airway device shown in the image given below on a patient. Which among the following statements is true regarding the adjunct?



A It can cause damage to the teeth

B ✓ It can be used in conscious patients

C It does not cause laryngospasm

D It can be safely used in facial trauma cases

Correct Answer: It can be used in conscious patients

The image shows a **nasopharyngeal airway** and its insertion. The true statement is that it **can be used in conscious patients**.

A nasopharyngeal airway is used to provide **artificial airway passage** to the hypopharynx. To place a nasopharyngeal airway, the device should be **lubricated** and inserted into the nose with the **bevel facing the nasal septum**, at a **90-degree angle** to the **longitudinal axis** of the body. It **bypasses** the **gag reflex** as they do not go through the mouth. Thus, it can be used in conscious patients or those who have been lightly anesthetized.

Care should be taken to **avoid excessive force** as damage to the nasal mucosa can result in epistaxis. The **nasal approach** should be **avoided** in conditions that put patients at **higher risk** for **epistaxis**. These conditions include but are not limited to coagulopathy, pregnancy, hereditary hemorrhagic telangiectasia, and uncontrolled hypertension. Absolute contraindications include basilar skull fractures, **facial trauma,** midface disruption, and injury to the roof of the mouth.

An oropharyngeal airway is used to maintain the airway in **anesthetized patients**. This adjunct prevents the tongue and epiglottis from falling back against the posterior pharynx and blocking the airway. They can cause coughing, retching, or **laryngospasm** in **conscious** patients.

To place an oropharyngeal airway, the device is inserted with the **curved portion** facing **downwards**. It is then rotated 180 degrees to be put in place. The below image shows the oropharyngeal airway:

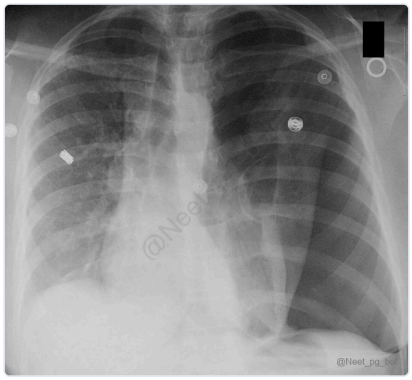


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References: Miller's Anesthesia 9th edition, Page 1388
Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK513220/

Q28. Anesthesia NEET 2024 ID: MG6765

In the following condition as shown in the radiograph, which of the following procedures would be difficult to perform? "



A Nasogastric tube insertion

B Laryngeal mask airway

C ✓ Tracheostomy

D Indirect laryngoscopy

Correct Answer: Tracheostomy

Tracheostomy would be difficult to perform in this case of pneumothorax with tracheal deviation.

The trachea is displaced from its usual midline position, making it challenging to **accurately locate and access** the trachea for the incision. The tracheostomy procedure requires precise placement to ensure proper airway management and avoid complications.

Steps of Tracheostomy:

- **Incision:** Make a vertical incision in the midline from the cricoid cartilage to above the sternal notch. For elective procedures, a 5 cm transverse incision two fingers' breadth above the sternal notch can be used for a better cosmetic outcome.
- **Dissection:** Dissect tissues in the midline. Displace or ligate any dilated veins encountered to control bleeding.
- **Muscle Separation**: Separate and retract the strap muscles of the neck laterally to expose underlying structures.
- **Thyroid Isthmus Management**: Displace the thyroid isthmus upwards or divide and ligate it with sutures.
- **Local Anesthesia**: Inject a few drops of 4% lignocaine into the trachea to suppress coughing.
- **Tracheal Incision**: Fix the trachea with a hook and make a vertical incision between the third and fourth tracheal rings, converting it into a circular opening. Avoid dividing the first tracheal ring to prevent perichondritis and stenosis.
- **Inserting the Tube**: Insert a tracheostomy tube of appropriate size into the tracheal opening and secure it with tapes.
- **Avoiding Tight Suturing**: Avoid tightly suturing or packing the skin incision to prevent subcutaneous emphysema.
- **Dressing:** Place a gauze dressing between the skin and the flange of the tracheostomy tube around the stoma to absorb secretions, provide a barrier against infection, and allow air to escape, promoting proper healing

Other Options:

Option A: Nasogastric tube insertion: Targets the esophagus, so tracheal deviation has minimal impact.

Option B: Laryngeal mask airway (LMA): Positioned over the larynx, not directly affected by tracheal deviation.

Option D: Indirect laryngoscopy: Focuses on the larynx and vocal cords, and is less influenced by the trachea's position.

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References: Bailey & Love's Short Practice of Surgery - 27th Edition, Page no: 744-749

Q29. Anesthesia NEET 2024 ID: MG6727

Which of the following agents is appropriate for reversing vecuronium when renal function tests are normal?

A Physostigmine

B Rivastigmine

C ✓ Sugammadex

D Succinyl choline

Correct Answer: Sugammadex

Sugammadex is used for immediate reversal of vecuronium-induced neuromuscular blockade/relaxant in the given scenario.

Sugammadex is a **modified γ-cyclodextrin** and the first selective relaxant–binding agent based on an **encapsulating principle** for the inactivation of nondepolarising neuromuscular blocking drugs mainly **rocuronium** and **vecuronium**. This principle of complex formation of sugammadex and rocuronium or vecuronium in a **1:1 ratio** occurs at **all levels of neuromuscular blockade** resulting in a more fast-acting pharmacologic reversal when compared with anticholinesterase drugs like neostigmine.

Other options:

Options A and B: **Physostigmine** and **Rivastigmine** are acetylcholinesterase inhibitors, which are **not** effective for reversing vecuronium.

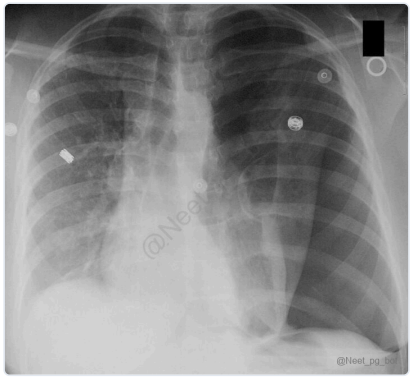
Option D: **Succinylcholine** is a neuromuscular blocker itself and would **not reverse vecuronium**; it is used to induce paralysis rather than reverse it.

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References: Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page 377,399,404,421
Miller's Anesthesia 9th edition, Page 851

Q30. Anesthesia NEET 2024 ID: MG7095

A COVID-19 patient admitted to the ICU on mechanical ventilation suddenly develops hypoxia. The patient has decreased breath sounds on the left side and increased intrathoracic pressure. What is the most likely diagnosis?



A ✓ Pneumothorax

B Left-Sided Pleural Effusion

C Acute Respiratory Distress Syndrome

D Left-Sided Pneumonia

Correct Answer: Pneumothorax

This patient in the above clinical scenario has developed a **sudden** onset of **hypoxia, decreased breath sounds** on one side, and **increased intrathoracic pressure,** along with an X-ray showing **hyper lucency** in the **left side** with **tracheal** and **mediastinal shift** to the **right** side is suggestive of a diagnosis of **Pneumothorax.**

A pneumothorax occurs when air enters the pleural space, causing **lung collapse** and preventing normal lung expansion, which leads to **absent** or **diminished breath sounds** over the affected area. Pneumothorax can cause a **rapid increase in intrathoracic pressure**, which impairs lung expansion and leads to significant hypoxia. In patients on mechanical ventilation, it delivers positive pressure to the lungs, which can exacerbate or reveal underlying pleural issues. Increased positive pressure can contribute to the development or worsening of a tension pneumothorax.

Other options:

Option B: Left-Sided Pleural Effusion: While this condition can also cause decreased breath sounds and hypoxia, it typically does not cause acute increases in intrathoracic pressure.

Option C: Acute Respiratory Distress Syndrome (ARDS): ARDS is characterized by diffuse lung inflammation and can lead to hypoxia. However, it generally does not cause localized decreased breath sounds.

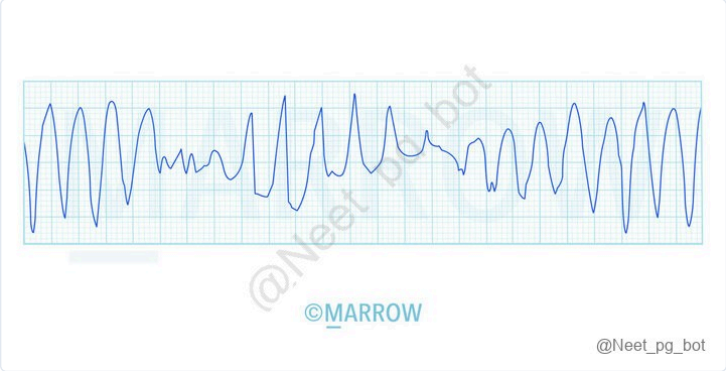
Option D: Left-Sided Pneumonia: Pneumonia can cause hypoxia and decreased breath sounds but is less likely to be associated with a sudden increase in intrathoracic pressure.

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References: Harrison's Principles of Internal Medicine, 20th Edition, Page no. 1998, 2008
Harrison's Principles of Internal Medicine, 21st Edition, page no: 2200

Q31. Anesthesia NEET 2024 ID: MG6726

A 3-year-old child presents with drowsiness and refusal to feed. On examination, the child is in a stuporous state. The ECG is shown below. What is the next best step in the management of this patient?



A Immediate synchronized cardioversion @ 1 joule/kg

B Defibrillate, check rhythm and then start CPR

C ✓ Defibrillate followed by CPR for 2min, and check rhythm

D Stop CPR and do cardioversion at 2 joules/ kg

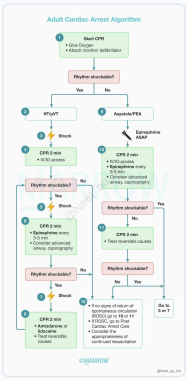
Correct Answer: Defibrillate followed by CPR for 2min, and check rhythm

The given ECG showing **polymorphic ventricular tachycardia**, which is a shockable rhythm, in a **stuporous** paediatric patient warrants **immediate defibrillation** followed by **CPR** for **2 minutes**.

Polymorphic ventricular tachycardia, especially in a paediatric patient, is often associated with severe underlying conditions and is a life-threatening rhythm disturbance. **Immediate defibrillation** is critical for restoring a normal rhythm.

After defibrillation, it is advisable to do CPR for 2 minutes before reassessing the rhythm for the next round of shock to ensure adequate blood circulation and oxygenation. .

Note: **Cardioversion** is not suitable for **polymorphic VT;** defibrillation is the preferred approach in such cases.



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References: Online Resource, https://cpr.heart.org/-/file:///D:/charan/anki/media/cpr-files/cpr-guidelines-files/highlights/hghlghts_2020_ecc_guidelines_english.pdf

Q32. Anesthesia NEET 2024 ID: MG7205

At what pressure does the marked structure open?



A ✓ 30-40 cm H2O

B 30-40 mm Hg

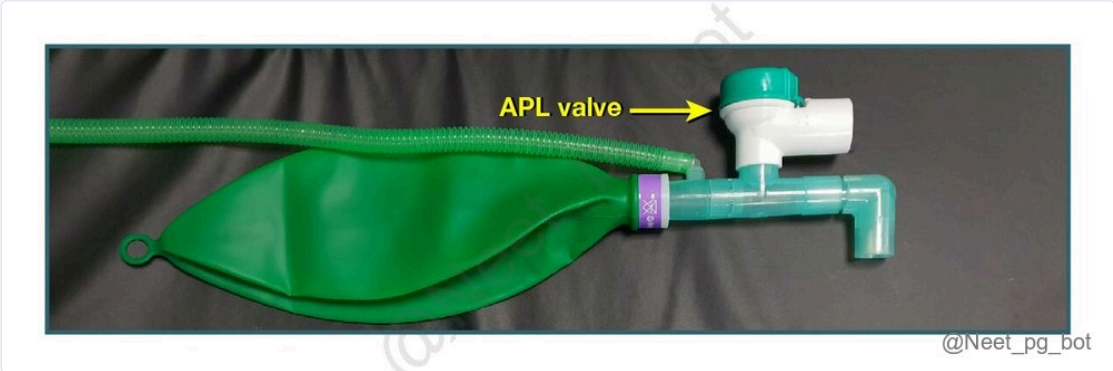
C 10-20 cm H2O

D 10-20 mm Hg

Correct Answer: 30-40 cm H2O

In the image, the marked structure is the **adjustable pressure-limiting (APL) valve** of Mapelson-C anaesthetic breathing system and it will open at the pressure of **30-40 cm H2O.** This valve is designed to release gas when a certain pressure is exceeded to prevent excessive pressure from building up in the patient's airway.

The APL valve typically opens at pressures of **30-40 cm H_2O** to allow for safe ventilation without causing barotrauma to the lungs. This range is appropriate for manual ventilation to prevent high airway pressures that could lead to lung injury.

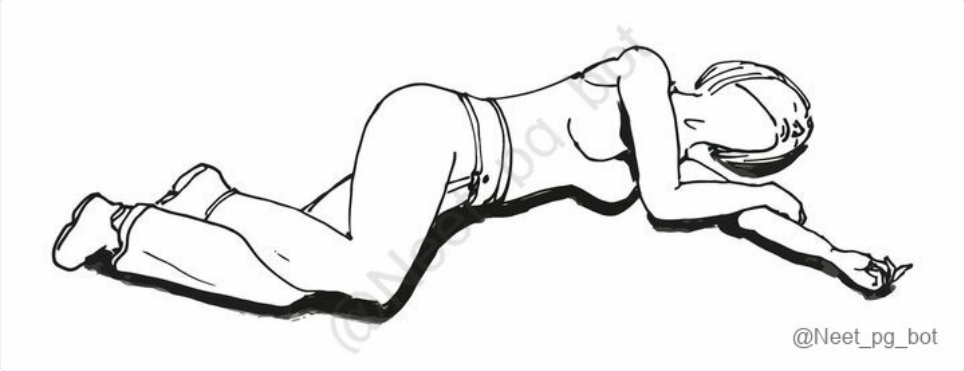


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References: Equipment in Anaesthesia and Critical Care, Page No: 84
Miller's Anesthesia 9th edition, Page No: 603, 604

Q33. Anesthesia NEET 2024 ID: MG7191

This recovery position is appropriate for:



A Unresponsive patient, pulse and breathing absent

B Responsive patient, pulse and breathing present

C ✓ **Unresponsive patient, pulse and breating present**

D Unresponsive with pulse, without breathing

Correct Answer: Unresponsive patient, pulse and breating present

The position illustrated in the picture is known as the **recovery position,** and it is used in the management of patients who are **unresponsive** but with **pulse present** and **breathing present.**

The recovery position can help **prevent complications** such as **aspiration** and **airway obstruction** in **unconscious** individuals. Additionally, it is crucial to **continuously monitor** the patient's **airway, breathing**, and **circulation** while in recovery and seek **immediate medical assistance**. An unresponsive patient who is breathing spontaneously in the supine position is at risk of airway obstruction from their tongue, mucus, vomit, etc. A **modified lateral position** is used and the following 6 principles are considered while choosing which variation of recovery position to use,

1. The victim should be in as near a true **lateral position** as possible, with the head dependent to allow free drainage of fluid.
2. The position should be stable.
3. **Avoid any pressure** on the chest that impairs breathing.
4. It should be possible to turn the victim on his or her side and to return to the back easily and safely, with concern for a possible cervical spine injury.
5. Good observation of and access to the airway should be possible.
6. The position itself should not cause injury to the victim.

Other options:

Option A: Unresponsive with pulse and breathing absent - In this situation, the patient is in cardiac arrest and requires **immediate cardiopulmonary resuscitation (CPR)**, not the recovery position. CPR involves chest compressions and rescue breaths to support circulation and respiration.

Option B: Responsive patient with pulse and breathing present- The patient should be monitored until the emergency response team arrives.

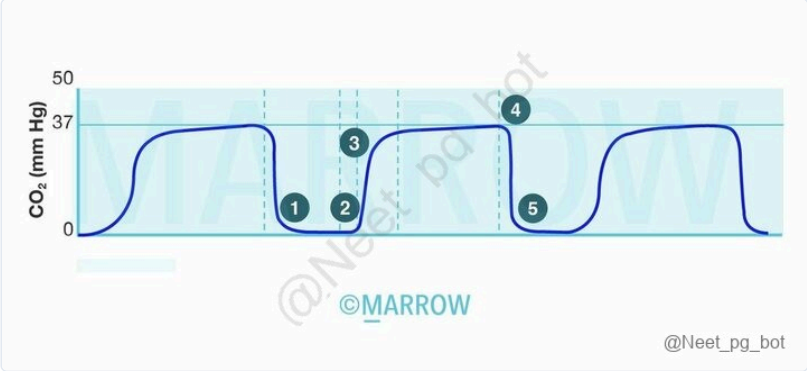
Option D: Unresponsive with pulse present and breathing absent - In this case, the patient is **experiencing respiratory arrest**. **Rescue breathing** or **bag-valve-mask ventilation** should be initiated to support the patient's breathing. The recovery position is not indicated in this scenario.

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References: Online Resource, https://www.ahajournals.org/doi/full/10.1161/circ.102.suppl_1.I-22#d3e1950
Online Resource, https://www.nhs.uk/conditions/first-aid/recovery-position/

Q34. Anesthesia NEET 2024 ID: MG7185

Following is the image of a normal capnograph. What is true about the segment 3-4?



A Inhalation of fresh gas

B Exhalation of dead space air

C Exhalation of mixed gases

D ✓ Alveolar gas and plateau pressure

Correct Answer: Alveolar gas and plateau pressure

In a capnographic waveform, the plateau phase corresponds to **alveolar air and plateau pressure.**

The normal waveform has four different phases:

Phase I: **Inspiratory baseline**, represents the CO_2-free gas from the airways (anatomical and apparatus dead space).

Phase II: **Expiratory upstroke,** mixing of dead space gas with alveolar gas.

Phase III: **Alveolar plateau**, represents CO_2-rich gas from the alveoli. The last of the alveolar gas is sampled to measure $ETCO_2$.

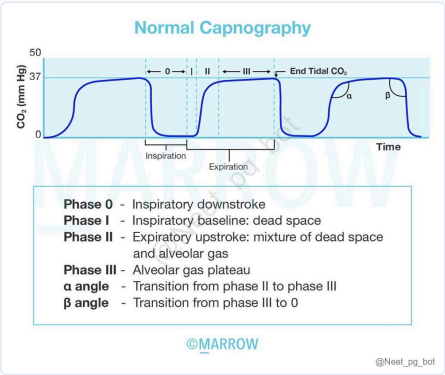
Phase 0: **Inspiratory downstroke** and the beginning of inspiration.

Alpha angle: The transition from phase II to III.

The alpha angle can be used to assess the ventilation/perfusion of the lung. V/Q mismatches will have an alpha angle greater than 90 degrees.

Beta angle: The transition from phase III to 0.

The beta angle can be used to assess rebreathing. If rebreathing occurs, the angle is greater than 90 degrees.

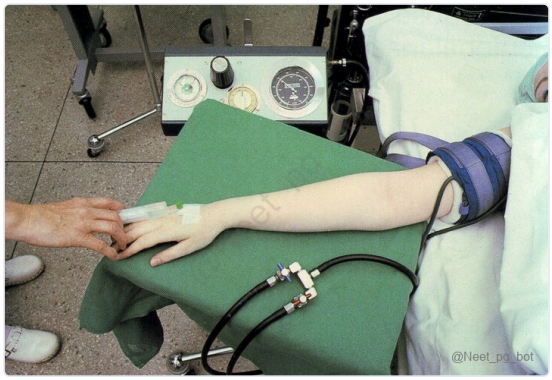


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References: Miller's Anesthesia 8th edition, Page nos 1551 and 1553
Miller's Anesthesia 9th edition, Page no. 1309

Q35. Anesthesia NEET 2024 ID: MG7184

What is the procedure being done in the image?



A ✓ Bier Block

B Axillary nerve block

C Median nerve block

D Wrist block

Correct Answer: Bier Block

The position in the image of a hand with a tourniquet occluded limb shows a **Bier Block.**

IVRA or **Bier's block** involves the intravenous administration of a **local anesthetic** into a **tourniquet-occluded limb.** It is used to provide anesthesia for short surgical procedures (<90 minutes) on the limbs.

Other options:

Option B: The **axillary nerve** block provides anesthesia for **short surgical procedures** on the hand, wrist, or forearm. It is performed by using a ultrasound-guided needle in the **proximal axilla** between the **tendon** of the **lattisimus dorsi** and the **teres major.**

Option C: **Median nerve** blocks are used for procedures involving the **palmar aspect** of the **arm** or the **carpal tunnel.** For a median nerve block at the elbow, the **probe** is placed in the **elbow crease** over the **proximal radius** and the anesthetic is injected aiming for the **fascial** plane between **flexor digitorum superficialis** and **flexor digitorum profundus.**

Option D: For a **wrist block,** the arm is abducted with the hand in **supination,** following which the **anesthetic** is **infiltrated** in the **dorsal aspect** of the wrist. This is used for short procedures of the wrist and the fingers.

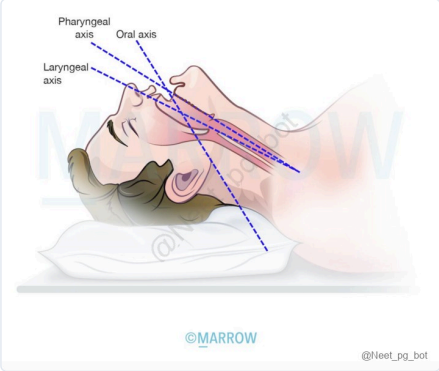
Note: Bupivacaine is not used for Biers block. It is the most cardiotoxic local anesthetic. Accidental early deflation of the tourniquet can lead to massive systemic doses of bupivacaine resulting in cardiac arrhythmias and cardiovascular collapse.

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References: Miller's Anesthesia 8th edition, Page no: 1041, 1042
Miller's Anesthesia 9th edition, page 1463
Online Resource, Pester JM, Bechmann S, Varacallo M. Median Nerve Block Techniques. [Updated 2023 Aug 4]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK459141/
Online Resource, https://www.bjaed.org/article/S1743-1816(17)30497-3/pdf

Q36. Anesthesia NEET 2024 ID: MG7183

What is true about the following position?



A High rate of error

B Ideal position for bougie insertion in emergencies

C Ideal postion for neurosurgery

D ✓ Sniffing position for intubation

Correct Answer: Sniffing position for intubation

The above image shows the **sniffing position** used for **intubation,** which is the true statement.

The sniffing position is a **common head** and **neck alignment** technique used during intubation to optimize the view of the glottis for easier insertion of the endotracheal tube. It is achieved by **flexing** the **neck** and **extending** the **head,** resembling the **posture** one assumes when **sniffing.** The position aligns the **oral, pharyngeal,** and **laryngeal** axes to provide a **straight line** of **sight,** making the **glottis** more **visible** during direct laryngoscopy and intubation.

Steps of intubation using the sniffing position:

- **Direct laryngoscopy** is performed in the sniffing position, with the head elevated 5-10 cm above the surgical table, the **cervical spine flexed,** and the **atlantooccipital joint extended.**
- The epiglottis is lifted up by the tip during laryngoscopy with **Miller's** blade.
- While performing laryngoscopy, the scope is lifted upwards and never levered over the upper incisors.

Other options:

Option A: The sniffing position is the most **commonly** used **position** in **anesthesia** as it ensures patient safety by ensuring a straight line of sight, allowing for better outcomes of the procedure.

Option B: The sniffing position is not the ideal position for **bougie insertion** during **emergencies** as there is concerns for **cervical spine trauma** in patients in the emergency department.

Option C: The sniffing position is not the typical position for neurosurgery. **Depending** on the location of the **surgery, different head** and **neck** positions are used.

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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no: 325 and 331
Miller's Anesthesia 9th edition, page no: 2125

Q37. Anesthesia NEET 2024 ID: MG7161

A patient presents with stridor after prolonged intubation. The endoscopic view is shown in the image below. What is the staging system for this condition?



A ✓ Cotton Meyer's

B AJCC

C Radkowski

D TNM

Correct Answer: Cotton Meyer's

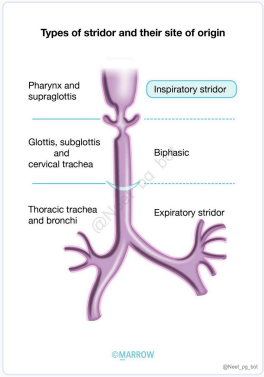
This patient presenting with **stridor** after **prolonged intubation** most likely has **subglottic stenosis post intubation.** This condition is staged by the **Cotton Meyer's** grading.

Subglottic stenosis following **prolonged intubation** is a narrowing of the airway just **below** the **vocal cords** (subglottic region), which can lead to difficulty breathing. This condition is often a complication of endotracheal intubation due to the **pressure** exerted by the **tube** or **cuff** on the **airway mucosa,** leading to **injury** and **scarring.**

These patients present with **stridor, dyspnea, hoarseness** or changes in voice, and **difficulty** in **breathing.**

Diagnosis is by **laryngoscopy** or **bronchoscopy** which allows **direct visualization** of the **subglottic area** to assess the degree of stenosis.
CT or **MRI** may be used to evaluate the **extent** of **narrowing** and identify any underlying structural damage.

Treatment can be via **endoscopic balloon** or **mechanical dilation,** laser therapy or surgical resection may be needed to remove the stenosis in severe cases. In severe cases, a **tracheostomy** may be required to **bypass** the **obstructed segment** and allow for breathing.



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References: P L Dhingra Diseases of Ear Nose and Throat and Head and neck surgery 7th edition, Page no 333-335
Cummings Otolaryngology: Head and Neck Surgery, Page no 983

Q38. Anesthesia NEET 2024 ID: MG2659

A patient is undergoing implant surgery for his upper limb. On giving general anaesthesia he develops tachycardia, sudden raise in body temperature and rigidity of muscles. What drug is most likely to have caused this?

A Propofol

B Etomidate

C Nitrous oxide

D ✓ Halothane

Correct Answer: Halothane

In the clinical scenario given above, the development of **tachycardia** and **muscle rigidity** soon after administration of **anesthetic** agents can be due to **malignant hyperthermia**. **Halothane** is the anesthetic agent associated with this condition.

Malignant hyperthermia is a rare genetic hypermetabolic disease of the muscle. It occurs due to a mutation of the gene for the ryanodine receptor (**Ryr 1**), located on chromosome 19. It is an ion channel responsible for calcium release from the **sarcoplasmic reticulum**.

The symptoms develop when the patient is exposed to **suxamethonium/succinylcholine** or other **inhaled anesthetics** like **halothane, isoflurane**, **sevoflurane, desflurane**, etc. The mutated ryanodine receptor leads to a sudden release of calcium from the sarcoplasmic reticulum of the skeletal muscle. This uncontrolled increase in intracellular calcium removes the inhibition of troponin, resulting in **sustained muscle contraction**, and **increasing adenosine triphosphatase (ATPase) activity**. This results in an uncontrolled hypermetabolic state with **increased oxygen consumption** and **CO2 production**, producing **severe lactic acidosis** and hyperthermia.

The key clinical features are:

- Succinylcholine-induced masseter muscle rigidity (causing trismus) or generalized muscle rigidity
- Tachycardia
- Hypercarbia (increased CO2 production) and reduced oxygen saturation
- Hyperthermia (rise of 1° Celsius can occur every 5 minutes)
- Arrhythmia
- Hypertension
- **Myoglobinuria** and **myoglobinemia** indicating muscle damage
- Metabolic derangements like **hyperkalemia** and mixed metabolic and respiratory acidosis

Treatment is to **discontinue the triggering agents** i.e. succinylcholine followed by administration of **100% oxygen**. Metabolic acidosis and hyperkalemia need to be corrected. Hyperthermia must be controlled by cooling the patient. **Dantrolene sodium** is the drug of choice. It is given at a dose of 2.5 mg/ kg intravenously and can be repeated every 5-10 minutes till the attack subsides.

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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page 1185-1188

Q39. Anesthesia NEET 2019 ID: MC3996

What is the mechanism of action of the curare group of muscle relaxants?

A Persistently depolarizing at neuromuscular junction

B ✓ Competitively blocking the binding of ACh to its receptors

C Repetitive stimulation of ACh receptors on muscle endplate

D Inhibiting the calcium channels on presynaptic membrane

Correct Answer: Competitively blocking the binding of ACh to its receptors

The **curare** group of drugs includes **non-depolarizing** neuromuscular blocking agents that act by **competitively blocking** the **binding** of **acetylcholine** to its receptors.

This leads to a decrease in the effect of acetylcholine and thus, neuronal transmission to the muscle does not occur.

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References: Miller's Anesthesia 8th edition, Page nos 959, 960, and 965

Q40. Anesthesia NEET 2019 ID: MC6383

Which among the following IV anaesthetics does not cause myocardial depression?

A ✓ Etomidate

B Propofol

C Thiopentone

D Ketamine

Correct Answer: Etomidate

Etomidate, a short-acting I.V. anesthetic agent doesn't cause myocardial depression.

Etomidate is the most **cardiac stable** intravenous agent due to its lack of effect on the sympathetic nervous system and on the function of the baroreceptor. Hence, etomidate is the I.V. agent of choice in patients with **coronary artery disease, cardiomyopathy,** and **cerebral vascular disease.**

Side effects of Etomidate:

- The highest incidence of nausea and vomiting among inducing agents
- Superficial thrombophlebitis and pain on injection
- Adrenocortical suppression

Note: Although Ketamine has an indirect sympathetic stimulatory effect, it possesses a **direct negative inotropic effect** on the myocardium. Hence, ketamine has to be used cautiously in critically ill patients with shock, as they are prone to have a depletion of endogenous catecholamines. Ketamine infusion in such patients can unmask its direct negative inotropic effects and further decrease the blood pressure and cardiac output.

Hence, for this MCQ, Etomidate is the best answer.

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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no 185
Miller's Anesthesia 8th edition, Page no 854

Q41. Anesthesia NEET 2019 ID: MA6552

Which of the following is not used for induction of anaesthesia in pediatric patients?

A Halothane

B Sevoflurane

C ✓ Desflurane

D Nitrous oxide

Correct Answer: Desflurane

Desflurane is a **highly pungent gas** and can produce **laryngospasm** if used for inhalational induction. Hence it is not used as an induction agent.

The incidence of **laryngospasm** following desflurane induction is almost **50%**. It is usually used for **maintenance of anaesthesia** only and is of particular value in obese patients, neurosurgical and spinal fusion procedures where **rapid awakening** is required for neurological function assessment.

Option A: **Halothane** is hepatotoxic but can be used (Sevoflurane is preferred)

Option B: **Sevoflurane** is the preferred inhalational agent for induction in the pediatric age group. Cardiovascular depression, bradycardia, and arrhythmias are less frequent with sevoflurane than with halothane and hence the former is preferred.

Option D: **Nitrous oxide** has the highest minimum alveolar concentration and so it is the least potent. It is used in combination with other potent inhaled anaesthetics to reduce their dose. It can be used to hasten the induction of anaesthesia along with other induction agents.

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References: Miller's Anesthesia 8th edition, Page nos 2766 and 2767
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no 166
Miller's Anesthesia 9th edition, Page no. 1269

Q42. Anesthesia NEET 2019 ID: MA5113

Which of the following agents does not elicit pain on injection?

A ✓ Ketamine

B Propofol

C Etomidate

D Thiopentone

Correct Answer: Ketamine

Ketamine does not elicit pain on injection, instead produces profound **analgesia**.

Propofol is formulated as a lipid emulsion, which causes pain on injection. A new aqueous prodrug, **fospropofol** is associated with **less pain** on injection.

Etomidate, although it causes pain on injection, the lipid formulation of etomidate is associated with a much less frequent incidence of pain.

Thiopentone at a concentration greater than 2.5% can produce pain on injection and venous thrombosis.

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References: Miller's Anesthesia 8th edition, Page nos 831, 851, and 854
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no 176

Q43. Anesthesia NEET 2019 ID: MA2757

Which among the following is most commonly used to monitor the depth of anaesthesia?

A EEG

B Provoked lower oesophageal contractility

C Entropy

D ✓ Bispectral index

Correct Answer: Bispectral index

The bispectral index is the most commonly used method to monitor the depth of anesthesia.

Note**: Middle latency auditory evoked potential** is the most sensitive indicator to evaluate the depth of anesthesia.

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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no 130
Online Resource, http://www.ijaweb.org/article.asp?issn=0019-5049;year=2007;volume=51;issue=5;spage=365;epage=365;aulast=Sinha#ft51
Morgan and Mikhail's Clinical Anesthesiology, 7th edition, page 235

Q44. Anesthesia NEET 2019 ID: MD0301

Which of the following is measured by the device shown in the given image?



A ✓ Oxygen saturation

B Oxygen content of the blood

C Partial pressure of oxygen

D Amount of inspired oxygen

Correct Answer: Oxygen saturation

The given image depicts a **pulse oximeter**. Pulse oximetry is used for **monitoring pulse rate and oxygen saturation** of a patient.

A small sensor is placed on the finger, toe, earlobe, or another convenient place. The **relative absorption of light by oxyhemoglobin** (HbO2) and deoxyhemoglobin is assessed by **measuring the amount of red and infrared light emerging from tissues** traversed by light rays and processed by the device, producing an oxygen saturation level.

Pulse oximetry does not measure the absolute content or the partial pressures of oxygen or carbon dioxide.

A pulse oximeter is not always reliable as it reads:

- Falsely high - carboxyhemoglobinemia
- Falsely low - severe anemia, low cardiac output states, hypovolemia, increased systemic vascular resistance
- 80% to 85%, irrespective of the arterial oxygen saturation - methemoglobinemia

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References: Atls Student Course Manual: Advanced Trauma Life Support - 10th Edition, Page no 11

Q45. Anesthesia AIIMS 2017 ID: MA0663

A 40 year old female with burns over her abdomen, both limbs and back and presented after 8 hours. Which of the following is the formula for calculation of fluid infusion?

A ✓ 2 ml/kg X %TBSA for the next 16 hours

B 5 ml/kg X %TBSA for the next 16 hours

C 4 ml/kg X %TBSA for first 8 hours then 2 ml/kg X %TBSA for next 16 hours

D 4 ml/kg X%TBSA for the next 16 hours

Correct Answer: 2 ml/kg X %TBSA for the next 16 hours

Fluid resuscitation in burns:

According to ATLS 10th edition, 2018, the **resuscitation formula** used in thermal burns is **2 mL/kg × % TBSA.**

First 8 hours: 1 mL/kg × % TBSA (lactated Ringer solution)

Next 16 hours: 1 mL/kg × % TBSA (lactated Ringer solution)

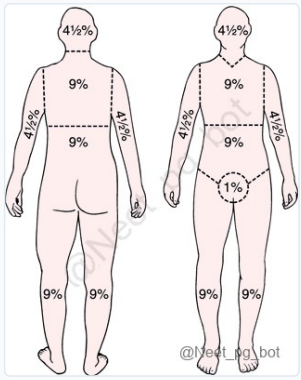
For a patient presenting 8 hours after sustaining thermal burns, the total volume using resuscitation formula should be given in the next 16 hours i.e; **2 mL/kg × % TBSA in the next 16 hours** while maintaining target urine output of 0.5 mL/kg/hr for adults.

In adults, urine output should be maintained between 30 and 50 cc/hr to minimize potential over-resuscitation.

Note:

The goal of resuscitation is to maintain the fine balance of adequate perfusion as indicated by urine output. It is important to understand that formulas provide a starting target rate; subsequently, the amount of fluids provided should be adjusted based on a urine output target of 0.5 mL/kg/hr for adults and 1 mL/kg/hr for children weighing less than 30 kg.

Total percentage of the burnt area is calculated clinically using the rule of nines.



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References: Advanced Trauma Life Support ATLS Student Course Manual 2018, Page no 173
Bailey & Love's Short Practice of Surgery - 27th Edition, Page no 624

Q46. Anesthesia AIIMS 2017 ID: MA0662

A 35-year-old woman was brought to the casualty ward with a history of road traffic accident (RTA) and head trauma. Her eye opens to pain, moans, can localize limb towards the pain stimulus on the left hand and away from the pain stimulus on the right hand and both the legs in extended posture. Calculate her GCS.

A 11

B ✓ 9

C 7

D 3

Correct Answer: 9

The GCS (Glasgow Coma Scale) of this patient is E2V2M5, which comes to a total score of 9.
E2-Eye opens to pain
V2-Moans
M5-Localizes to pain on the left hand (the score represents the best performance elicited, which would be localizing towards the pain stimulus)
Revised Glasgow coma scale for head injury:

Minimum GCS is 3, maximum GCS is 15.

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References: Advanced Trauma Life Support ATLS Student Course Manual 2018, Page no. - 110

Q47. Anesthesia AIIMS 2017 ID: MB9198

A patient who was on ventilator and being ventilated for past few days suddenly pulls out of the endotracheal tube. What is the next step of management?

A Re intubate

B Oxygenate and reintubate

C Start CPR

D ✓ Watch for spontaneous breaths and reintubate if needed

Correct Answer: Watch for spontaneous breaths and reintubate if needed

About half of self-extubated patients may not require re-intubation. Pre-extubation FiO2, PaO2/FiO2, and ventilator mode, as well as gender, may be useful in determining the likelihood that a self-extubated patient will require re-intubation and also watch for spontaneous breathing.

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References: Online Resource, https://www.ncbi.nlm.nih.gov/pubmed/11000731

Q48. Anesthesia AIIMS 2017 ID: MB9124

The time required for preoxygenation before tracheal intubation?

A 1 min

B ✓ 3 min

C 5-7 min

D 5 min

Correct Answer: 3 min

Preoxygenation with **tidal volume** breathing of **3 mins** is required before tracheal intubation.

Preoxygenation or **denitrogenation** is the process of replacing nitrogen in the lungs with oxygen.

This delays the onset of critical hypoxia during the period of apnea by **5-8 minutes.**

This lengthened apnea time provides an improved margin of safety while the anesthesiologist secures the airway and resumes ventilation.

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References: Journal, https://www.ncbi.nlm.nih.gov/pubmed/6402951
Miller's Anesthesia 8th edition, Page no 1653

Q49. Anesthesia AIIMS 2017 ID: MB9167

Nerve fibre least susceptible to local anesthetic is:

A A-delta

B A- gamma

C A- beta

D ✓ C

Correct Answer: C

The nerve fibre **least susceptible** to local anaesthetic is **type C**. The order of **susceptibility** of nerve fibres to local anaesthetics is **B >A-δ > A-γ > A-β > A-α > C** i.e

- **B fibres** are the **most susceptible/least resistant**
- **C fibres** are the **least susceptible/most resistant**

Factors that increase the sensitivity of nerve fibres are :

- Smaller diameter
- Unmyelinated nerves
- Anatomical and physiological factors

*Note: Although theoretically, C fibres are supposed to be the most susceptible to local anaesthetics, practically it was found to be the least susceptible.

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References: Morgan and Mikhail's Clinical Anesthesiology, 6th edition
Miller's Anesthesia 9th edition, Page no. 869, 1417
Ganong's Review Of Medical physiology, 26th Edition, page no 91,
Katzung Basic and Clinical Pharmacology 14th Edition, page no 465
Online Resource, https://www.uptodate.com/contents/clinical-use-of-local-anesthetics-in-anesthesia?search=nerve%20fibre%20susceptibility%20&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1#H2368018785

Q50. Anesthesia AIIMS 2017 ID: MB9224

Which of the following can be used as an intravenous induction agent?

A ✓ Lorazepam

B Bupivacaine

C Dexmedetomidine

D Neostigmine

Correct Answer: Lorazepam

Lorazepam is a benzodiazepine that can be used for **induction of anesthesia** at a dose of 0.1mg/kg, although **midazolam** is **preferred** among benzodiazepines.

Benzodiazepines are mainly used for **sedation** and less commonly for induction of anesthesia due to the availability of better induction agents (propofol, etomidate, ketamine, and thiopentone).

Dexmedetomidine is a selective alpha-2 agonist and is used primarily as an adjuvant. So, a better option would be lorazepam.

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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no, 179, 180, 288
Katzung's Basic and Clinical Pharmacology 12/E (Lange Basic Science), Page no 443

Q51. Anesthesia AIIMS 2017 ID: MD7393

Which of the following is the correct sequence?

A ✓ Assess the patient's response, Activate ERS, check carotid pulse, Start CPR

B Check carotid pulse, Start CPR, Call for help and activate ERS, Defibrillate

C Defibrillate, Assess response, Check carotid pulse, Maintain airway

D Start CPR, Activate ERS, Defibrillate, Check pulse

Correct Answer: Assess the patient's response, Activate ERS, check carotid pulse, Start CPR

BLS sequence:

First step – Assess safety of the surroundings.

Second step - Assess for responsiveness

Third step – Proceed if the patient is unresponsive. Shout for help.

If the patient is unresponsive assess for breathing and carotid pulse simultaneously for no more than 10 seconds.

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References: Harrison's Principles Of Internal Medicine - 19th Edition, Page no 1768
Online Resource, https://www.acls-pals-bls.com/algorithms/bls/

Q52. Anesthesia INI-CET 2024 ID: MG5793

Which of the following is a 2nd generation LMA device?

LMA - Laryngeal mask airway

A Intubating LMA

B ✓ Proseal LMA

C Flexible LMA

D Ambu LMA

Correct Answer: Proseal LMA

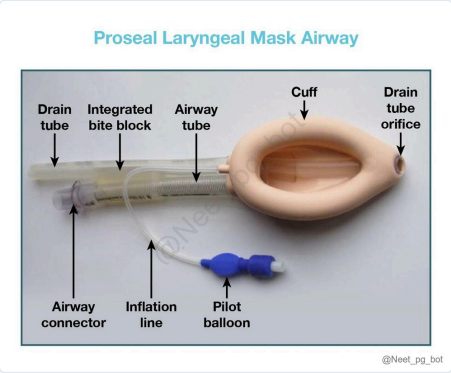
Proseal LMA is a 2nd generation LMA device as it has a **separate gastric tube** to decompress the stomach and **prevent gastric aspiration** associated with the classical LMA or 1st generation LMA. Other 2nd generation LMA's include **LMA Supreme**, **I-Gel** and **LMA Protector**.

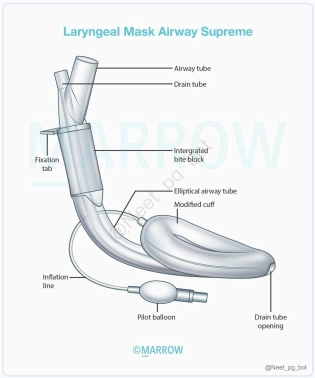
All other options are examples of 1st generation LMA's since they do not have a separate gastric tube to aspirate for gastric secretions.

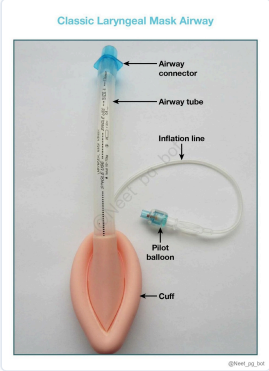
Flexible LMA allows the tube **complete flexibility** and resistance to compression so that the head and neck can be turned without dislodging the mask.

The **intubating LMA (ILMA)** and **Ambu LMA(ALMA)** are variants of the **LMA Classic**, a 1st generation LMA used to **aid in intubation** of the patient in difficult airways.

Note : Although Ambu LMAs are produced as 1st generation and 2nd generation LMA, Proseal is exclusively a 2nd generation LMA and hence the best answer to this question.







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References: Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page 1389-1391
Miller's Anesthesia 9th edition, Page 586

Q53. Anesthesia INI-CET 2024 ID: MG5801

A patient was planned for surgery under general anaesthesia. During induction of anaesthesia, rocuronium 85 mg was given and the patient was tried to be intubated. It resulted in a failed intubation. Which of the following drugs can be given for immediate reversal of the drug?

A Neostigmine

B Glycopyrrolate

C ✓ Sugammadex

D Edrophonium

Correct Answer: Sugammadex

Sugammadex is used for immediate reversal of rocuronium induced neuromuscular blockade/relaxant in the given scenario.(Option C)

Sugammadex is a **modified γ-cyclodextrin** and the first selective relaxant–binding agent based on an **encapsulating principle** for inactivation of nondepolarising neuromuscular blocking drugs mainly **rocuronium** and **vecuronium**. This principle of complex formation of sugammadex and rocuronium or vecuronium in a **1:1 ratio** occurs at **all levels of neuromuscular blockade** resulting in a more fast-acting pharmacologic reversal when compared with anticholinesterase drugs like neostigmine.

The stability of the rocuronium-sugammadex complex is a result of **Van der Waals forces**, **hydrogen bonds** and **hydrophobic interactions**. There is some binding affinity with pancuronium, but this interaction is too low to have a significant clinical effect.

Other options:

Option A: **Neostigmine** is used to reverse **depolarising** neuromuscular blockade such as **succinylcholine**, especially in the **phase II block**(fade of the train-of-four response) and sufficient time has passed for the circulating concentration of succinylcholine to be negligible. Therefore, neostigmine is slow-acting compared to sugammadex but the agent preferred in the case of succinylcholine.

Neostigmine administration during **phase I block, increases neuromuscular Ach** concentration and **inhibits pseudocholinesterase** (plasma cholinesterase) activity that can prolong the action of succinylcholine, hence it is given after some time during the phase II block.

Option B: To prevent **cholinergic side-effects** of neostigmine, **glycopyrrolate** is co-administered in a **1:5 ratio** to **neostigmine**(0.2mg: 1 mg). Atropine is an alternative but glycopyrrolate is preferred due to the incidence of tachycardia associated with atropine.

Option D: **Edrophonium** is a short-acting **acetylcholinesterase inhibitor** preferred in the case of **mivacurium** reversal since it does not show plasma cholinesterase inhibition present in neostigmine that can prolong mivacurium action.

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References: Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page 377,399,404,421
Miller's Anesthesia 9th edition, Page 851

Q54. Anesthesia INI-CET 2024 ID: MG5792

Which of the following are components of high-quality cardio-pulmonary resuscitation(CPR) in adults?

1. Chest compressions should be given at the rate of 100-120/min
2. The depth of compression should be 3-4 cm
3. Compression: breaths given should be in the ratio of 30:2
4. Should allow for complete recoil of the chest

A 1,2 and 3

B ✓ 1,3 and 4

C 2,3 and 4

D 3 and 4 only

Correct Answer: 1,3 and 4

The depth of compression should be **at least 5cm** to ensure high-quality CPR in an adult cardiac arrest.

The foundation of CPR is **high-quality chest compressions**. Compressing the chest pumps blood **from the heart to the brain** and then to the rest of the body. Each time the compressions are stopped, the blood flow from the heart decreases significantly. Once resumed, it takes several compressions to bring the blood flow back up to the levels present before the interruption.

The **components** of high-quality chest compressions in adults are as follows :

- **Position victim** - Position the victim face up on a **firm, flat surface**. This allows compression of the heart to create adequate blood flow.
- **Compression-to-ventilation ratio** - **Single rescuers** should use the compression-to-ventilation ratio of **30 compressions to 2 breaths** when giving CPR to victims of any age.
- **Compression rate** - Compress at a rate of **100 to 120/min**. This rate is the same for compression in all cardiac arrest victims.
- **Compression depth** - Compress the chest **at least 5cm**. Compressing the chest more than 6 cm in adults may decrease the effectiveness of the compression and cause injuries. Using a CPR-quality feedback device can help reach the optimal depth.
- **Chest recoil** - Allow the chest to **recoil completely** after each compression. This allows blood to flow into the heart. To help ensure complete recoil, **avoid leaning** on the chest between compressions. Chest compression and chest reoil times should be about equal.
- **Interruptions in chest compressions** - Minimize interruptions in chest compression. the proportion of time that rescuers perform chest compressions during CPR is called chest compression fraction (CCF). A CCF of **80% or highe**r should be the goal.

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References: Miller's Anesthesia 9th edition, Page 2531, 2532
Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page 2380-2393
ACLS Manual 2020, Page no : 125

Q55. Anesthesia INI-CET 2024 ID: MG5791

What is the inhalational anaesthetic agent used for induction in children?

A Nitrous oxide

B Isoflurane

C ✓ Sevoflurane

D Desflurane

Correct Answer: Sevoflurane

Sevoflurane is the inhalational anaesthetic agent used for induction in children. (Option C)

It is the preferred agent due to its **non-pungent** sweet-smelling nature and **rapid increase** in alveolar concentration on administration. Its **low blood solubility** helps during emergence from the effects of anaesthesia, reducing the post-operative delay in the patient's recovery from anaesthesia.

Desflurane and **isoflurane** are **avoided** for inhalation induction because they are **pungent** and associated with **coughing**, breath-holding and **laryngospasm**. They are usually used for the **maintenance of anaesthesia**. Desflurane is preferred in patients with **renal** and **liver disease** and isoflurane is preferred in **neurosurgery** and **cardiac surgery.**

Due to its **second-gas effect**, **nitrous oxide** is mainly used as a **carrier gas** for other inhalational agents and not given alone.

@Neet_pg_bot

References: Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page 25, 312

Q56. Anesthesia INI-CET 2024 ID: MG7924

Ambu Aura 40 is an example of?

A ✓ Supraglottic laryngeal airway generation 1

B Supraglottic laryngeal airway generation 2

C Supraglottic laryngeal airway generation 3

D Supraglottic laryngeal airway generation 4

Correct Answer: Supraglottic laryngeal airway generation 1

The Ambu Aura 40 is an example of a **first-generation supraglottic airway device.**

Supraglottic airway (SGA) devices are classified into different generations based on their features and advancements:

First-generation SGA devices: Basic devices designed to provide a seal around the glottis for ventilation. They lack additional safety features, such as protection against aspiration. Examples include the original Laryngeal Mask Airway (LMA Classic) and Ambu Aura 40 (can be reused up to 40 times),

Second-generation SGA devices: These are improved devices with additional features to enhance safety and functionality. They typically have:
A gastric access channel (to allow gastric decompression or reduce the risk of regurgitation and aspiration).
Improved sealing mechanisms. Examples: Ambu Aura Gain (with drain tube), LMA ProSeal, and i-gel.

Third-generation and Fourth-generation devices: These are not standard classifications in clinical practice yet. However, devices beyond the second generation may feature advanced integration with monitoring systems, adjustable seals, or further innovations.

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References: Online Resource, https://pmc.ncbi.nlm.nih.gov/articles/PMC8023114/
Online Resource, https://www.uptodate.com/contents/supraglottic-airways-sgas-for-airway-management-for-anesthesia-in-adults?search=ambu%20aura&sectionRank=1&usage_type=default&anchor=H2309126498&source=machineLearning&selectedTitle=1%7E147&display_rank=1#H2309126498

Q57. Anesthesia INI-CET 2024 ID: MG7915

Which of the following drugs causes adrenal suppression?

A Thiopental

B Ketamine

C ✓ Etomidate

D Propofol

Correct Answer: Etomidate

Etomidate causes adrenal suppression which can last for 72 hours.

Etomidate inhibits **11β-hydroxylase**, resulting in decreased biosynthesis of cortisol and mineralocorticoids and increases the formation of intermediaries like 11-deoxycorticosterone.

It is a more potent **inhibitor of steroid synthesis** than as a sedative-hypnotic as the concentration which produces adrenocortical suppression (10 ng/mL)is much lower than that needed for hypnosis (>200 ng/mL).

This suppression is particularly concerning in critically ill patients, where stress-induced cortisol production is vital for survival.

Other Options:

Option A: Thiopental: Primarily causes **sedation and hypnosis.**

Option B: Ketamine: Does not cause adrenal suppression. Instead, it may slightly **stimulate adrenal catecholamine** release due to its action on the sympathetic nervous system.

Option D: Propofol: Can cause **CNS depression** without direct impact on adrenal hormone synthesis.

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References: Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page no: 319

Q58. Anesthesia INI-CET 2024 ID: MG7916

Which of the following is incorrect regarding thiopentone?

A It is lipid soluble

B It achieves maximum concentration in the brain after an IV infusion

C Redistribution to adipose tissue and muscle is seen

D ✓ Rapid decline in its effect is due to renal excretion

Correct Answer: Rapid decline in its effect is due to renal excretion

The **rapid decline** in the effect of thiopentone is due to its **rapid redistribution**, not due to renal excretion.

Thiopentone has a rapid onset of action because of its lipophilic nature, so it rapidly crosses the blood-brain barrier and enters highly perfused organs like the brain, heart, liver, and kidneys.

As thiopentone **redistributes** from the brain to tissues: it perfuses to **muscle, fat, and skin.** Over time, due to its **high lipid solubility**, thiopentone accumulates maximally in **fat**. Fat acts as a storage reservoir since it has low blood flow, and drug release from fat occurs slowly.

This **redistribution** reduces the concentration of thiopentone, **terminating** its anesthetic **effect.**

Other Options:

Option A: It is lipid-soluble: Thiopentone is highly lipid soluble, which allows it to cross the blood-brain barrier quickly and achieve rapid onset of action in the CNS.

Option B: It achieves maximum concentration in the brain after an IV infusion: Due to its lipophilicity, thiopentone rapidly accumulates in the brain after IV administration.

Option C: Redistribution to adipose tissue and muscle is seen: Thiopentone's effect is terminated by redistribution from the brain to adipose tissue and muscle. This accounts for the short duration of its anesthetic effect.

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References: Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page no: 322

Q59. Anesthesia INI-CET 2024 ID: MG7911

A female following an RTA injured her chest. On X-ray, a fracture of 3 to 5th ribs was noted. Pain was not responsive to analgesics. Which block will be given to reduce the pain?

A Cervical block

B ✓ Thoracic epidural block

C Supraclavicular block

D Lumbar epidural block

Correct Answer: Thoracic epidural block

A **thoracic epidural block** will be given to reduce the rib fracture pain.

Epidural anaesthesia is used to provide adequate analgesia which is the mainstay of treatment in patients with multiple rib fractures.

The **complications** can be minimized by adequate pain relief with epidural anaesthesia. Rib fracture patients will have the following complications due to splinting and pain:

- Hypoventilation
- Hypoxia

Other **techniques** to provide analgesia following rib fractures are **intercostal nerve block** and **intrapleural block** which necessitates multiple injections, which increases the risk for complications.

The **complications** include vascular puncture, pneumothorax, patient discomfort due to the need for palpation of the fractured rib during local anaesthetic (LA) administration. Also, LA toxicity can occur because of the high systemic absorption.

Rib fractures can cause severe pain that does not respond well to conventional analgesics. A thoracic epidural block is effective for pain relief in this area because it provides segmental anesthesia to the thoracic dermatomes and surrounding structures, helping reduce pain and improve respiratory function.

Indications of thoracic epidural block:

- Postoperative pain relief: After thoracic or upper abdominal surgeries.
- Trauma management: Rib fractures to reduce pain and improve ventilation.
- Chronic pain syndromes: Postherpetic neuralgia or cancer-related pain.

Other Options:

Option A: Cervical block: Used primarily for pain management or anesthesia involving the **neck and upper shoulders.**

Option C: Supraclavicular block: Used for anesthesia of the upper limb by blocking the **brachial plexus.**

Option D: Lumbar epidural block: Used for lower body anesthesia, including the **pelvis and lower extremities.**

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References: Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page no 1947

Q60. Anesthesia INI-CET 2024 ID: MG8031

Which of the following will lead to a decrease in diastolic BP?

A Systemic infusion of alpha-adrenergic blocker

B Systemic infusion of smooth muscle relaxant

C ✓ Loss of lumbar sympathetic outflow

D Systemic infusion of beta-1 adrenergic agonist

Correct Answer: Loss of lumbar sympathetic outflow

Loss of lumbar sympathetic outflow will lead to a decrease in diastolic blood pressure (DBP).

Loss or blockade of the lumbar sympathetic chain (**spinal anaesthesia**) causes dilation of capacitance (venous) vessels. This leads to a decrease in diastolic blood pressure and subsequent orthostatic lightheadedness or syncope.

Vasodilation is more marked on the **venous** than on the arterial side of the circulation, resulting in blood pooling in the venous capacitance vessels. This reduction in circulating blood volume is well tolerated at low levels of spinal anaesthesia in healthy patients. With an increasing level of block, this effect becomes more marked, and venous return becomes gravity-dependent. If venous return decreases too much, cardiac output and organ perfusion decline sharply.

Other options:

Option A: Systemic infusion of **alpha-adrenergic blocker** will minimally decrease diastolic BP, not significantly like in loss of lumbar sympathetic outflow.

Option C: Systemic infusion of **smooth muscle relaxants** like Sildenafil, which is a PDE 5 inhibitor (increases cGMP), does not cause a decrease in DBP.

Option D: Systemic infusion of **beta-1 adrenergic agonist** increases systolic BP and heart rate.

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References: Goodman & Gillman's The Pharmacological Basis Of Therapeutics - 14th Edition, Page no: 500

Q61. Anesthesia INI-CET 2024 ID: MG7914

Which of the following is true regarding paediatric cardiopulmonary resuscitation?

A Chest compression at the rate of 30:2 with 2 rescuers

B Adenosine via interosseous is preferred over other methods

C ✓ In chest compression, depth must be at least 1/3rd of the child's chest AP diameter

D Adrenaline given in dose 0.01 mg/kg iv/io in 1:1000 concentration

Correct Answer: In chest compression, depth must be at least 1/3rd of the child's chest AP diameter

The **depth of compressions** in infants should be at least 1.5 inches or **1/3 the depth of the infant's chest.**

Effective chest compressions in infants and pediatric population:

- 100 to 120 compressions per minute (30 compressions for every 15-18 sec).
- One provider: Place two fingers, one between the nipple line and the other 1 cm below the first finger
- Two providers: Encircle the infant's torso with both hands, thumbs pointing cephalad and positioned 1cm below the nipples over the sternum.
- Depth of compressions should be at least 1.5 inches or 1/3 the depth of the infant's chest.
- Press hard and fast
- Allow for full chest recoil after every compression

Other Options:

Option A: Chest compression at the rate of 30:2 with 2 rescuers: For children, the compression-to-ventilation ratio is **15:2 when there are two rescuers**. A **30:2 ratio** is used only if there is a **single rescuer**, similar to adult CPR.

Option B: Adenosine via interosseous is preferred over other methods: Adenosine is not used in pediatric cardiopulmonary resuscitation. It is used in **PSVT** treatment in **intravenous routes.**

Option D: Adrenaline given in dose 0.01 mg/kg iv/io in 1:1000 concentration: Epinephrine is the drug of choice for treating bradycardia in infants and children. The recommended dose is **0.01 mg/ kg of a 1:10,000 solution.**



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References: Online Resource, Online: https://www.acls-pals-bls.com/algorithms/bls/

Q62. Anesthesia INI-CET 2024 ID: MG7912

Which of the following must be avoided in post-cardiac arrest resuscitation care?

A Maintain MAP over 65

B Maintain SpO2 above 94%

C Revascularisation procedure if the patient develops ST elevation post-resuscitation

D ✓ Bring the body temperature down to below 32 degrees, therapeutic hypothermia if unresponsive

Correct Answer: Bring the body temperature down to below 32 degrees, therapeutic hypothermia if unresponsive

The body temperature in post cardiac arrest care is to be maintained between **32-36 degrees** for at least **24 hours**. This is known as targeted temperature managament.

Targeted temperature management (TTM) is the only intervention demonstrated to improve neurological recovery after cardiac arrest and involves maintaining core body temperature between 32-36°C for at least 24 hours.

This is done in post-cardiac resuscitation patients who have achieved a return of spontaneous circulation (ROSC) but are still **comatose.**

The entire body is to be **cooled** using a combination of methods like fluid-filled cooling blankets, ice packs, forced air blankets, cold IV fluids, and/or invasive devices or catheters.

It is the only intervention that can improve neurological recovery after cardiac arrest. It is most beneficial in ventricular fibrillation and pulseless ventricular tachycardia patients.

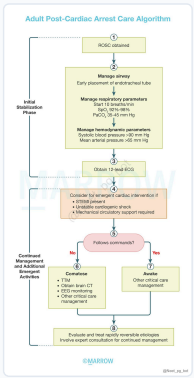
Cardiac arrest (especially with shockable rhythms) and **neonatal hypoxic-ischemic encephalopathy** are the primary established indications for therapeutic hypothermia. **Stroke** (in some cases), where thrombectomy is not an option, is another indication.

Other Options:

Option A: Maintain MAP over 65: A MAP of ≥ 65 mmHg is the minimum threshold recommended during post-cardiac arrest care to prevent ischemic injury and support organ function.

Option B: Maintain SpO2 above 94%: The recommended target SpO_2 range in post-cardiac arrest care is 92-98%.

Option D: Revascularisation procedure if the patient develops ST elevation post resuscitation: Post-cardiac arrest patients with STEMI benefit from revascularization, as it reduces mortality and improves cardiac function.



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References: Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page no: 2416
Online Resource, https://pubmed.ncbi.nlm.nih.gov/16221780/
Online Resource, https://pubmed.ncbi.nlm.nih.gov/20354545/

Q63. Anesthesia INI-CET 2024 ID: MG7902

Jackson Rees circuit is used in

- A ✓ Paediatric anaesthesia
- B Cardiac anaesthesia
- C Neuro anaesthesia
- D Obstetric anaesthesia

Correct Answer: Paediatric anaesthesia

The Jackson Rees circuit is used in **paediatric anaesthesia.**

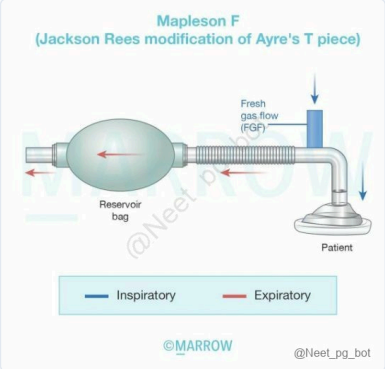
Jackson Rees modification of Ayre's T piece is classified as a Mapleson's F system (**Semi-Closed Circuits**).

It is used in children <6 years or <20 kgs.

Mapleson Circuit	Usage
Type A (Magill's circuit)	Circuit of choice for spontaneous ventilation
Type B	Obsolete
Type C (Water's to and fro)	Obsolete
Type D (Bain's circuit)	Circuit of choice for controlled ventilation. Most commonly used circuit.
Type E (Ayre's T piece)	Circuit of 2nd choice in children
Type F (Jackson Rees Modification)	Circuit of choice in children (<20 kg)

Other options:

Option B, C, and D: **Cardiac anaesthesia, Neuro anaesthesia, and Obstetric anaesthesia** require **closed circuit.** They have a provision to absorb CO_2, using soda lime. Closed circuits are used for **long periods** of procedures.



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References: Miller's Anesthesia 9th edition, Page No: 611-614

Q64. Anesthesia INI-CET 2021 ID: MF0927

Ventilatory support is required if SpO2 becomes less than 90% because

A ✓ Tissues become vulnerable for hypoxia below 90%.

B Peripheral chemoreceptors becomes insensitive to hypoxia below 90%.

C Central chemoreceptors becomes insensitive to hypoxia below 90%.

D Patient capacity to generate adequate tidal volume decreases below 90%.

Correct Answer: Tissues become vulnerable for hypoxia below 90%.

Tissues become vulnerable to **hypoxia** when SpO2 falls below 90% and thus supplementary oxygen therapy is indicated below this oxygen saturation level.

In adults, children, and infants (older than 1 month), When SpO2 falls below 90%, the PaO2 becomes less than **60 mmHg** and hence to prevent tissues from hypoxia, supplemental oxygen therapy should be started.

In neonates, when SpO2 falls below 88%, the PaO2 becomes less than 50 mmmHg and hence supplemental oxygen therapy should be initiated in these patients.

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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no: 1281

Q65. Anesthesia INI-CET 2021 ID: MF0883

In conventional oxygen therapy, which device will deliver highest FiO2?

A Venturi mask

B Hudson mask

C ✓ Non rebreathing mask

D Nasal cannula

Correct Answer: Non rebreathing mask

Non rebreathing masks provide highest FiO2 during conventional oxygen therapy.

The non-rebreather mask is a **low flow device** that can be used to **deliver oxygen** at **higher concentrations** (FiO2 80-95%). It uses a **reservoir bag** to let the oxygen get concentrated while a **one-way valve** doesn't allow to breathe in expired air.

This type of masks are used in patients with significant hypoxia, with relatively normal ventilation, such as victims of trauma or carbon monoxide exposure.

Other options:

A) Venturi mask:

- Venturi mask is a **high-flow, fixed-performance** oxygen delivery system and the highest oxygen concentration (**FiO_2**) delivered is **60% at a flow rate of 15 L/min.**
- It is based on the **Bernoulli Principle**, which states that in order to maintain a constant flow, a fluid must increase its velocity as it flows through a constriction.
- It is available in different colors, with the green valve venturi mask providing 60% FiO_2.
- This is useful with patients who require greater FiO_2 such as **COPD** patients, where it is important to provide controlled oxygenation.

B) Hudson mask:

- Also known as **simple face mask**
- Low-flow, variable performance device
- They provide an oxygen flow rate of up to **5-10 L/min** and provide a variable FiO2.
- The body of the mask functions as reservoir for both oxygen and expired carbon dioxide.
- Used in patients who require oxygen therapy for shorter periods of time.

D) Nasal cannula:

- They are low-flow, variable performance devices, with flow rate of **1-4 L/min.**
- Used in patients who require low supplemental oxygen.
- Nasopharynx acts as the reservoir in such cannulas.

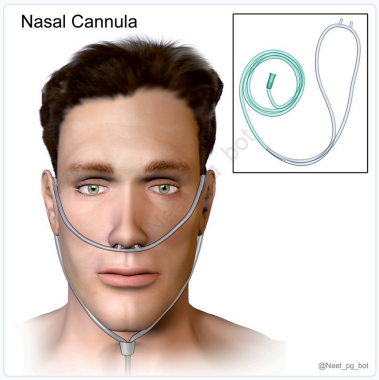
Indications of a nasal cannula and other low-flow oxygen delivery systems include

- Minute ventilation < 8–10 L/min
- Breathing frequencies < 20 breaths/min
- Normal inspiratory flow (10–30 L/min).









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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no: 1282- 1285
Equipment in Anaesthesia and Critical Care, Page no: 17, 18, 20, 21
Online Resource, https://openanesthesiajournal.com/VOLUME/11/PAGE/35/FULLTEXT/

Q66. Anesthesia INI-CET 2021 ID: MF0898

What is the nerve blocked in the following local anesthesia technique?



A ✓ Nasociliary nerve

B Anterior ethmoidal nerve

C Sphenopalatine nerve

D Greater palatine nerve

Correct Answer: Nasociliary nerve

The given image shows a **nasociliary nerve block.**

The procedure is done by inserting the needle 1 cm superior and 1.5 cm posterolateral to the medial canthal ligament of the eye.

The nasociliary nerve is a branch of the ophthalmic nerve which in turn is a branch of the trigeminal nerve.

Branches:

- **Long ciliary nerve-** cornea, ciliary body, iris
- **Ramus communicans** to the ciliary ganglion
- **Posterior ethmoidal nerve**- sensory innervation to frontal sinus and posterior ethmoidal air cells
- **Anterior ethmoidal nerve**- sensory innervation to middle and anterior ethmoidal cells as well as lateral part of the nasal septum
- **Infra-trochlear nerve** (continuation of the nasociliary nerve)- provides innervation to the conjunctiva, lacrimal sac, caruncle, the medial part of the eyelids

Uses of nasociliary nerve block:

Used in

- Surgery of nasal fractures,
- Septoplasty,
- Rhinoplasty and
- Repair of skin laceration.

The above image represents an anterior ethmoidal nerve block.





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References: Online Resource, https://web.archive.org/web/20200212135121id_/https://www.jorapain.com/doi/pdf/10.5005/jp-journals-10046-0131
Online Resource, https://books.google.co.in/books?hl=en&lr=&id=cd-on7Rx0dYC&oi=fnd&pg=PA22&dq=nasociliary+nerve+block&ots=w40qWZBGR4&sig=fCQ2M0rf2InISH8KsUoUa45IwME&redir_esc=y#v=onepage&q=nasociliary%20nerve%20block&f=false
Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK551696/

Q67. Anesthesia INI-CET 2021 ID: MF0881

How long should elective surgery be deferred for a patient who had cough and fever, after he has recovered from COVID-19, and had not required any hospitalization?

A ✓ 6 weeks

B 8-10 weeks

C 12 weeks

D 4 weeks

Correct Answer: 6 weeks

A patient, who has recovered from COVID-19 and hadn't required **any hospitalization** should be deferred from any elective surgery for a period of **6 weeks.**

The American Society of Anesthesiologists has suggested the following duration of intervals for patients recovered from COVID-19. The wait time is from the date of diagnosis of COVID-19 to the date of surgery as follows:

- 4 weeks: for an asymptomatic patient or recovery from only mild, non-respiratory symptoms.
- 6 weeks: for a symptomatic patient (e.g: cough, dyspnea) who did not require hospitalization
- 8-10 weeks: for a symptomatic patient who is hospitalized or who has comorbidities such as diabetes or who is immunocompromised.
- 12 weeks: for a patient who was admitted in intensive care unit due to COVID-19 infection.

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References: Online Resource, https://www.asahq.org/in-the-spotlight/coronavirus-covid-19-information/elective-surgery

Q68. Anesthesia INI-CET 2021 ID: MD7821

A 42-year-old man who was under home isolation after testing positive for COVID-19, develops breathing difficulty and is brought to the ER. The patient is alert and responsive, but is breathing rapidly. SpO2 is measured to be 86%. He is administered oxygen via a face mask with a reservoir bag with a flow rate of 10 L/min. After a trial of one hour, his SpO2 is 88%. What is the next step in management?

A Increase flow rate to 15 L/min

B ✓ High-flow nasal cannula oxygen therapy

C Endotracheal intubation

D No specific intervention necessary

Correct Answer: High-flow nasal cannula oxygen therapy

This patient has **persistent hypoxia** after an adequate trial of standard oxygen therapy and should be started on **high-flow nasal cannula oxygen therapy.**

When **respiratory distress** and/or **hypoxemia** of the patient cannot be alleviated after receiving standard oxygen therapy, high–flow nasal cannula oxygen therapy (**HFNO**), or **non–invasive ventilation** can be considered. Compared to standard oxygen therapy, HFNO reduces the need for intubation.

Patients on HFNO should be closely monitored, and **endotracheal intubation** should be done if the patient acutely deteriorates or does not improve after a short trial (about 1 hr).

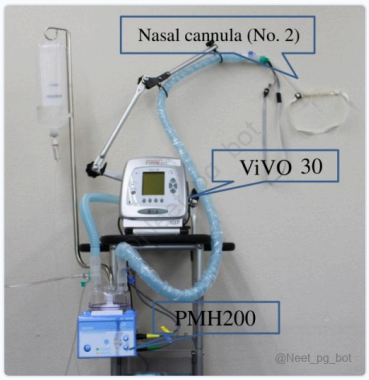
Patients with hypercapnia (exacerbation of COPD), hemodynamic instability, multi-organ failure, or abnormal mental status should not receive HFNO.

Note: If a high flow nasal cannula (HFNC) or simple nasal cannula is used to administer O2, an N95 mask should be applied over it.

Some of the benefits of HFNC are

- It can deliver oxygen at **40L/min**, 37°C, and 99.9% relative humidity.
- The **high-flow oxygen** is delivered directly to the nasopharynx. This produces **FiO_2** equal to that delivered by traditional mask devices.
- It also **reduces the work** of breathing, **improves gas exchange** by nasopharyngeal dead-space washout, and **decreases energy** required to **humidify** and **heat** respiratory gases.

The HFNC equipment shown in the image below uses a heated humidifier (named PMH200), a nasal cannula (no. 2), and a ventilator (brand: vivo30).



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References: Online Resource, NATIONAL CLINICAL MANAGEMENT PROTOCOL COVID-19, Government of India, Version 5 03.07.20
https://www.mohfw.gov.in/pdf/UpdatedClinicalManagementProtocolforCOVID19dated03072020.pdf
Miller's Anesthesia 9th edition, Page no 2595

Q69. Anesthesia INI-CET 2021 ID: MF0889

What is the mechanism of action of local anesthetics?

- A ✓ Block sodium channels
- B Block potassium channels
- C Block calcium channels
- D Block chloride channels

Correct Answer: Block sodium channels

Local anesthetics primarily act by blocking the **sodium channels** bound to the membrane.

Sodium channels have one larger α subunit and one or two smaller β subunits. These voltage-gated sodium channels exist in 3 states :

- Resting (non conducting)
- Open (conducting) and
- Inactivated (non conducting)

The local anesthetics bind to the α subunit of sodium channels, thereby preventing their activation and inhibiting the influx of sodium.

Local anesthetics have a higher affinity for the sodium channels in open and inactivated state than in the resting state.

Local anesthetic inhibition is both voltage and frequency-dependent and is higher when nerve fibers are firing rapidly than with slow, infrequent depolarizations.

Note:

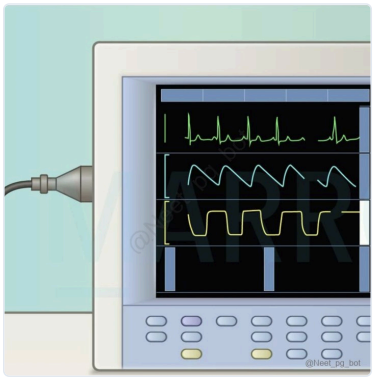
Local anesthetics may also bind and inhibit calcium, potassium, transient receptor potential vanilloid 1 (TRPV1) and many other channels and receptors. But blocking of sodium channels is the best answer, of all the given options.

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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no: 264- 266

Q70. Anesthesia INI-CET 2021 ID: MF0905

You are in the operating room and notice the tracing in yellow color on this device. What does it indicate?



A ✓ Capnography

B Airway pressure

C O2 pressure in inhaled air

D O2 pressure in exhaled air

Correct Answer: Capnography

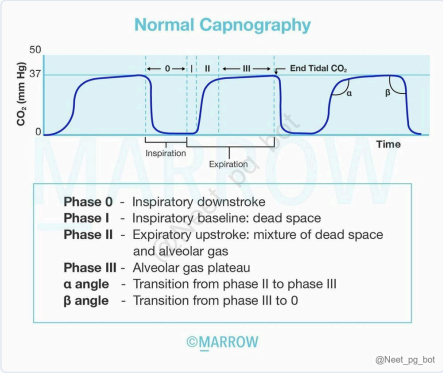
The reading marked as X in the monitor is the **capnography**.

Waveform capnography is a continuous **non-invasive** measurement and graphical display of the **EtCO2**.

It is a direct measure of **lung ventilation** and an indirect measure of **circulation** and **metabolism**.

The **normal capnograph** displays the **exhaled CO_2** in both inspiratory and expiratory phases. The expiratory phase is divided into three distinct components:

- **Phase I**: exhalation of dead space gas (i.e., partial pressure of CO_2, pCO_2 ~0).
- **Phase II**: A sharp **rise** in **pCO_2** to a **plateau**indicates the sampling of transitional gas between the airways and alveoli.
- **Phase III**: The plateau region of the capnograph, corresponds to the pCO_2 in the alveolar compartment.
- **Following phase III,** a sharp inspiratory downstroke occurs, which is referred to as the beginning of phase 0 or phase IV by some.



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References: Miller's Anesthesia 8th edition, Page no: 1552- 1554

Q71. Anesthesia INI-CET 2021 ID: MF0875

What is the most common cause for hospital admission after day care surgery?

A Respiratory distress

B Post surgical pain

C ✓ Postoperative nausea and vomiting

D Bleeding

Correct Answer: Postoperative nausea and vomiting

Postoperative nausea and vomiting (PONV) is the most common cause of unanticipated hospital admission following daycare surgery.

The two **most common complications** of daycare surgery are **PONV** and **postoperative pain**. PONV refers to the presentation of **symptoms** in the postoperative care unit or **within 24 hours** of the procedure. Pain is usually managed preoperatively and postoperatively with the administration of analgesics. While it is a common complication, **postoperative pain** is **not** the **most common** cause of **unanticipated hospital admission**. Opioid medications which are commonly used to treat postoperative pain present with side effects such as nausea and vomiting which may aggravate PONV.

The occurrence of PONV can be based on the following factors, patient, choice of anesthetic agent, and the procedure. The risk factors are:

- Patient: **Women** (higher incidence in pregnancy), history of PONV or motion sickness, nonsmokers, **younger age** (incidence decreases as the age increases).
- Anesthetic agent: **General anesthesia**, inhalational agents (isoflurane, sevoflurane), **nitrous oxide**, **longer duration** of anesthesia, preoperative opioid administration.
- Procedure: Laparoscopic procedures, lithotripsy, gynecologic procedures, adenotonsillectomy, strabismus surgery.

The drugs used in the prophylaxis and **treatment** of PONV include **ondansetron**, dexamethasone, and **transdermal scopolamine**.

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References: Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page no: 1772
Online Resource, https://www.uptodate.com/contents/office-based-anesthesia?search=ambulatory%20surgery&source=search_result&selectedTitle=1%7E150&usage_type=default&display_rank=1#H1108389999
Online Resource, https://www.uptodate.com/contents/postoperative-nausea-and-vomiting?search=postoperative%20nausea%20and%20vomiting&source=search_result&selectedTitle=1%7E109&usage_type=default&display_rank=1#H959820332

Q72. Anesthesia INI-CET 2021 ID: MF2635

What is the frequency of tetanic stimulation used in neuromuscular monitoring?

A 0.2 Hz

B 2 Hz

C 20 Hz

D ✓ 50 Hz

Correct Answer: 50 Hz

The commonly used **frequency** of **tetanic stimulation** in neuromuscular monitoring is **50 Hz**.

Tetanic stimulation is a method to evaluate residual neuromuscular blockade.

Tetanic stimulation consists of a series of extremely rapid electrical stimuli typically applied over 5 seconds. The most commonly used frequency is 50 Hz stimulation given for 5 seconds.

- During a **depolarizing block**, the muscle response to 50 Hz tetanic stimulation for 5 seconds is **sustained**.
- In contrast, during a **phase II block** after the injection of succinylcholine, the response following tetanic stimulation is not sustained i.e., a **fade occurs.**

Disadvantages of tetanic stimulation are:

- This method has only 70% sensitivity to detect residual paralysis and its **specificity** is only about **50%**. Hence, this test is of limited value for assessing neuromuscular recovery.
- This method is very **painful**, which makes it difficult to use in unanesthetized patients.
- During the late phase of neuromuscular recovery, tetanic stimulation can produce **lasting antagonism of neuromuscular block** in the stimulated muscle. This may change the response of the tested site such that it may no longer be representative of other muscle groups.

Note:

Other patterns of nerve stimulations, which are used to assess the patients for residual blockade are

- **Train of four stimulation**: Here four supramaximal stimuli (**2 Hz**) are delivered every 0.5 seconds.
- **Double-burst stimulation**: Here two short bursts of **50 Hz** tetanic stimuli is delivered, which are separated by 750 ms.

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References: Miller's Anesthesia 9th edition, Page no. 835 and 1359

Q73. Anesthesia INI-CET 2021 ID: MF2155

A 60-year-old male patient was undergoing laparoscopic cholecystectomy. During the surgery, a sudden drop in EtCO2 to 8 mmHg was noted. His SpO2 became 90%, his blood pressure dropped to 80/50 mmHg, and peak airway pressure was 18 cm of H2O. What is the diagnosis?

1. CO_2 embolism
2. Severe bronchospasm
3. Pneumothorax
4. Single bronchus intubation

EtCO2- End-tidal carbon dioxide

A Both 1 and 2

B Both 3 and 4

C ✓ 1 only

D 2 only

Correct Answer: 1 only

A sudden fall in $EtCO_2$ with hypotension (fall in BP to 80/50 mmHg) along with normal airway pressure (< 35 to 40 cm of H_2O) during surgery is suggestive of **CO_2 embolism.**

Intraoperative pulmonary embolism usually presents as sudden cardiovascular collapse and hypoxemia with a **sudden fall in $EtCO_2$**. Transesophageal echocardiography (**TEE**) is the most sensitive device for detecting intracardiac air embolus. If air is identified in the right atrium, or if it is suspected, immediate central vein cannulation and aspiration of the air may be lifesaving.

The placement of a vena cava filter should be considered postoperatively.

The best method to confirm the position of the endotracheal tube is the **end-tidal CO_2** concentration assessed by **capnography.**

Other options:

Intraoperative bronchospasm is usually manifested as wheezing, **increasing peak airway pressure**, and a slowly rising waveform on the capnograph.

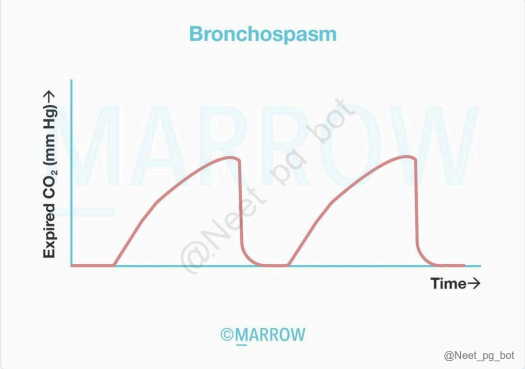
A **pneumothorax** may manifest as **hypercapnia** along with hypoxemia, hypotension, **increased peak airway pressure**, and decreasing tidal volumes. It may result in abrupt cardiovascular collapse unresponsive to fluid and vasopressor administration.

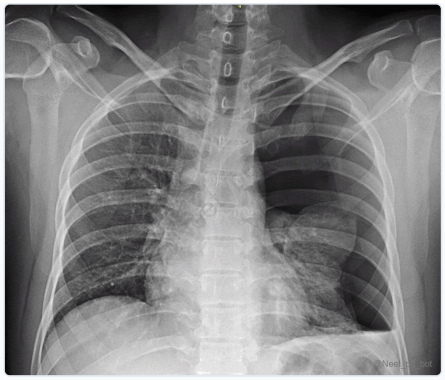
Deep sulcus sign on a supine chest radiograph points towards pneumothorax. The costophrenic angle is abnormally deepened when the pleural air collects laterally, producing the deep sulcus sign.

The below chest X-ray shows left pneumothorax. Notice the collapsed lung and absent lung markings on the left side. Also mediastinal shift towards the right side.

Endobronchial intubation shows **increased** peak expiratory pressure**,** hypoxemia, increased airway pressure, asymmetric chest expansion and the absence of breath sounds over one lung, generally the left.

Note: **Peak airway pressure** is the amount of airway pressure delivered by the ventilator to overcome the resistance of the endotracheal tube, airway, and alveoli.





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References: Miller's Anesthesia 9th edition, Page no. 1878
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Pg 333, 527,537,541
Online Resource, https://www.emra.org/emresident/article/demystifying_ventilator_alarms/

Q74. Anesthesia INI-CET 2021 ID: MF2153

The maximum oxygen flow rate which can be kept with a conventional oxygen mask is

A 5 L/min

B 15 L/min

C ✓ 10 L/min

D 20 L/min

Correct Answer: 10 L/min

The maximum oxygen flow rate which can be kept with a **conventional oxygen mask** is **10 L/min.**

A **conventional oxygen mask** is a simple face mask that is used routinely in hospitals. It covers the nose and mouth of the patient. There are 2 exhalation ports through which the patient exhales carbon dioxide. This mask is held in place with elastic around the back of the head and a metal piece that fits over the nose. It provides **40% to 60%** of O_2 concentration and delivers O_2 at the rate of **6 to 10 L/min.**

Conventional oxygen therapy includes **all the devices** that are used in routine practices. It includes nasal prongs, low-flow nasal cannula, or masks. It is the main supportive treatment administered to patients after planned extubation.

Oxygen delivery devices	Flow rate (liter/min)	Approximate FiO2
Nasal cannula	1-6	0.24-0.44
Simple face mask	6-10	0.40-0.60
Non-rebreathing mask	10-15	0.90-1.00
Venturi mask	2-15	0.24-0.60

Oxygen delivery devices can be broadly divided into low flow oxygen devices (eg: nasal cannula, simple face mask, non-rebreathing mask) and high flow oxygen devices (eg: venturi mask).

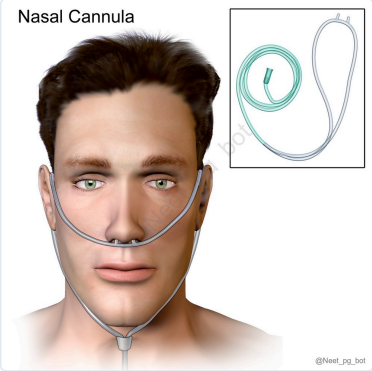
Important note- If the question is about the maximum flow rate with the conventional oxygen **mask**, it would be **10L/ min**. Whereas if the question is asking about the maximum oxygen flow rate with conventional oxygen **therapy**, then it would include all the conventionally used devices. Therefore, the answer would be **15L/min.**

The below image shows a **nasal cannula**, which delivers oxygen at **1-6 L/min.**

The below image shows a **non-rebreathing** mask that provides an oxygen flow rate of up to **10-15 L/min.**

The below image shows a **venturi mask**, which delivers oxygen at **2-15 L/min.**









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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no: 1282- 1285
Online Resource, https://openanesthesiajournal.com/contents/volumes/V11/TOATJ-11-35/TOATJ-11-35.pdf
Online Resource, https://ccforum.biomedcentral.com/articles/10.1186/s13054-019-2465-y
Online Resource, https://opentextbc.ca/clinicalskills/chapter/5-5-oxygen-therapy-systems/

Q75. Anesthesia INI-CET 2021 ID: MF2154

Which of the following can be used for induction of anesthesia, maintenance and can be safely used in endotracheal intubation?

A Halothane

B Desflurane

C ✓ Sevoflurane

D Isoflurane

Correct Answer: Sevoflurane

Among the given options, **sevoflurane** can be used for induction and maintenance of anesthesia. It can also be safely used in endotracheal intubation

Inhalation agents such as sevoflurane and halothane are useful in the induction of anesthesia, But sevoflurane due to its **nonpungency** and **rapid onset** makes it an excellent choice for smooth and rapid inhalation inductions in pediatric and adult patients. Sevoflurane has low blood solubility and partition coefficients resulting in more rapid emergence.

Sevoflurane also produces adequate muscle relaxation for intubation of children following an inhalation induction.

Other options:

Option A: **Halothane** can also be used for gaseous induction of anesthesia. Its disadvantage is long induction time owing to the high blood:gas partition coefficient. However, the incidence of halothane **hepatitis** and myocardial depression has reduced its use.

Option B: **Desflurane** produces **laryngospasm** when used for inhalational anesthesia and hence it is not used for induction. Laryngospasm is because of the **pungent** nature of the gas.

Option D: **Isoflurane** causes respiratory depression as well as upper respiratory irritation.

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References: Morgan and Mikhail's Clinical Anesthesiology, 6th edition, Page no 152, 168,169,170,172
Morgan and Mikhail's Clinical Anesthesiology, 7th edition, Page no 303 - 315

Q76. Anesthesia INI-CET 2021 ID: MF2156

During resuscitation, a patient received a 20% Intralipid injection. What could have been the most probable diagnosis?

A ✓ Local anaesthesia toxicity

B Asphxyia due to drowning

C Electrocution

D Pulseless electrical activity

Correct Answer: Local anaesthesia toxicity

Patients who are unresponsive due to **local anesthesia toxicity** require 20% Intralipid injection during resuscitation.

Accidental intravascular injection of local anesthetic like bupivacaine results in **bupivacaine-induced cardiotoxicity.**

Initial management includes immediate initiation of cardiopulmonary resuscitation (**CPR**) and **defibrillation** as per the advanced cardiac life support (**ACLS**) algorithm.

- Following CPR, **Intralipid therapy** is initiated. A rapid bolus of **Intralipid 20% IV** (1.5 mL/kg) is administered without delay, followed by an infusion of 0.25 mL/kg/min. The infusion should be given continuously for at least 10 minutes after hemodynamic stability is achieved.
- Intralipid decreases cardiac bupivacaine concentration by 11% within 3 minutes of administration, and cerebral bupivacaine concentration by 18% within 15 minutes.

Bupivacaine is a long-acting local anesthetic that is the most **cardiotoxic** among the local anesthetics. It depresses the rapid phase of depolarization in Purkinje fibers and ventricular muscle due to a decreased availability of fast sodium channels in cardiac membranes. This reduced rate of depolarization **depresses myocardial contractility.** Hence, bupivacaine is a potent cardio-depressant.

Note: Local anesthesia systemic toxicity (LAST) can also affect the central nervous system (CNS), which is more susceptible to local anesthetic action as compared to the cardiovascular system. Dizziness, tinnitus, and disorientation are the initial symptoms. Shivering, muscular twitching, and tremors follow. **Tonic-clonic seizures** occur. Finally, respiratory depression can lead to respiratory arrest.

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References: Miller's Anesthesia 9th edition, Page 885
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page 274
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6001420/

Q77. Anesthesia INI-CET 2021 ID: MF2376

A 50-year-old male patient has grade 3 ASA. Which of the following postbronchodilator PFT findings requires further evaluation of the lung function of the patient?

ASA - American Society of Anesthesiology
PFT - Pulmonary function test
FEV1 - Forced expiratory volume in 1 second
FVC - Forced vital capacity

A FEV1 < 0.85 and FEV1/FVC <0.8

B FEV1 < 0.85 and FEV1/FVC <0.75

C ✓ FEV1 < 0.8 and FEV1/FVC <0.7

D FEV1 < 0.8 and FEV1/FVC <0.75

Correct Answer: FEV1 < 0.8 and FEV1/FVC <0.7

FEV1 <0.8 and FEV1/FVC <0.7 are classified as **moderate COPD** under **GOLD classification** and they require further evaluation of the lung function.

Preoperative pulmonary function tests (PFTs) assess the severity of known pulmonary disease, diagnose causes of respiratory symptoms, and may help to identify patients who are at risk for postoperative pulmonary complications.

The severity of COPD is assessed by using the Global Initiative for Lung Disease (GOLD) criteria. In pulmonary function testing, a postbronchodilator FEV1/FVC ratio of <0.70 is considered diagnostic for COPD.

The Global Initiative for Chronic Obstructive Lung Disease (GOLD) system categorizes airflow limitation into stages. In patients with FEV1/FVC <0.70:

- GOLD 1: mild: FEV1 ≥80% predicted
- **GOLD 2: moderate: 50% ≤ FEV1 <80% predicted**
- GOLD 3: severe: 30% ≤ FEV1 <50% predicted
- GOLD 4: very severe: FEV1 <30% predicted.

This is important in the perioperative period as the diagnosis of COPD is an independent and significant risk factor in the development of postoperative pulmonary complications. Pre-operative FEV_1 and DLCO of >80% imply a lower risk for a permanent postoperative pulmonary disability. Below these values, further risk-benefit assessment of the surgery needs to be examined and more information should obtained.

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References: Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4613400/
Online Resource, https://bestpractice.bmj.com/topics/en-us/7/criteria
Online Resource, https://pubmed.ncbi.nlm.nih.gov/20662279/
Online Resource, https://www.uptodate.com/contents/chronic-obstructive-pulmonary-disease-definition-clinical-manifestations-diagnosis-and-staging?search=COPD&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1#H13

Q78. Anesthesia NEET 2021 ID: MF1352

A patient undergoing surgery is maintained on anesthesia with halothane. During the procedure, the patient suddenly developed hyperthermia and muscle rigidity. Which of the following drugs is most likely implicated in this condition?

A D-curare

B ✓ Suxamethonium

C Cis-atracurium

D Rocuronium

Correct Answer: Suxamethonium

The given clinical scenario of a patient being operated under halothane, developing **hyperthermia** and **muscle rigidity** is suggestive of an episode of **malignant hyperthermia.** The drug most likely implicating/causing this condition is **suxamethonium** (succinylcholine).

Suxamethonium is a peripherally acting **depolarizing muscle relaxant,** commonly administered during the induction of anesthesia to relax muscles of the jaw, neck, and airway and thereby facilitate laryngoscopy and endotracheal intubation.

Malignant hyperthermia is a rare genetic hypermetabolic muscle disease. It occurs due to mutation of the gene for the **ryanodine receptor** (Ryr 1), located on chromosome 19. Ryr 1 is an ion channel responsible for calcium release from the sarcoplasmic reticulum.

The symptoms develop when the patient is exposed to **inhaled anesthetics** (halothane in this case) or **suxamethonium/succinylcholine.**

The mutated Ryr 1 leads to a sudden release of calcium from the sarcoplasmic reticulum of the skeletal muscle. This **uncontrolled increase** in **intracellular calcium** removes the inhibition of troponin, resulting in **sustained muscle contraction**, which in turn, leads to an increase in adenosine triphosphatase (ATPase) activity. This results in an uncontrolled hypermetabolic state with **increased oxygen consumption** and **CO2 production**, producing severe lactic acidosis and hyperthermia.

The key clinical features are:

- **Succinylcholine-induced masseter muscle rigidity** (causing trismus) or generalized muscle rigidity
- Tachycardia
- Hypercarbia (increased CO2 production) and reduced oxygen saturation
- Hyperthermia (rise of 1° Celsius can occur every 5 minutes)
- Arrhythmia
- Hypertension
- Myoglobinuria and myoglobinemia indicating muscle damage
- Metabolic derangements like hyperkalemia and mixed metabolic and respiratory acidosis

Treatment is to **discontinue** the **triggering agents** i.e; inhaled anesthetics and succinylcholine followed by administration of **100% oxygen.** Metabolic acidosis and hyperkalemia need to be corrected. Hyperthermia must be controlled by **cooling** the **patient.**

Dantrolene sodium is the drug of choice. It is given at a dose of **2.5 mg/ kg intravenously** and can be repeated every 5-10 minutes till the attack subsides.

Other options:

D-curare or D-tubocurarine (option A), cis-atracurium (option C), and rocuronium (option D) are non-depolarizing muscle relaxants and are not implicated in malignant hyperthermia.

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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page 1185-1188
Goodman & Gillman's The Pharmacological Basis Of Therapeutics - 13th Edition, Page no 397

Q79. Anesthesia NEET 2021 ID: MF1362

A 30-year-old male patient was intubated for surgery by a final-year resident. Which of the following is the best method to confirm the position of the endotracheal tube?

A X-ray chest

B Auscultation

C ✓ End-tidal CO2 concentration

D Chest rise

Correct Answer: End-tidal CO2 concentration

The best method to confirm the position of the endotracheal tube is the **end-tidal CO2 concentration** assessed by **capnography.**

Capnography is a **non-invasive method of measurement** and **graphical display** of partial pressure of **carbon dioxide (CO_2)** in exhaled breath or the **end-tidal CO_2**. **Capnograph** is the device used for measurement and **capnogram** is the **CO2 waveform** that is displayed. It is the **most reliable indicator** of endotracheal tube placement.

The proper placement of the endotracheal tube is indicated by a **normal CO_2 waveform** for at least **three breaths.**

Other signs to confirm proper endotracheal intubation are:

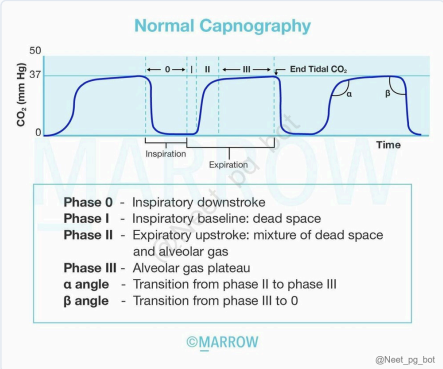
Symmetrical chest rise, presence of equal bilateral breath sounds, absent breath sounds over the epigastrium, visible condensation in the ETT, large exhaled tidal volume, and appropriate compliance of the reservoir bag during manual ventilation.

Absent **CO_2 waveform** tracing despite correct ETT intubation is seen in the following conditions:

- Severe bronchospasm,
- Equipment malfunction,
- Cardiac arrest, or hemodynamic collapse

Clinical relevance:

- Capnography does not differentiate **endotracheal** from **endobronchial** intubation. Signs of endobronchial intubation include increased peak expiratory pressure, hypoxemia, increased airway pressures, asymmetric chest expansion and the absence of breath sounds over one lung, generally the left.
- In **esophageal intubation**, the capnograph can initially detect some **CO_2** in the stomach from swallowing expired air. But after a few breaths, this **CO_2** will be washed out and **$ETCO_2$** will become zero.
- The most sensitive and earliest indicator of **malignant hyperthermia** is a persistent increase in **$ETCO_2$**.

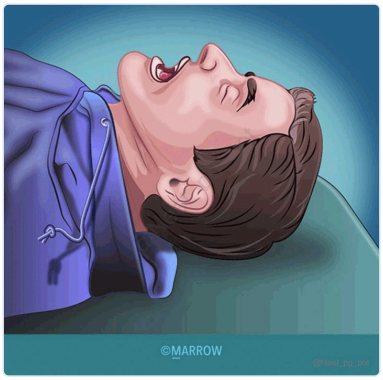


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References: Miller's Anesthesia 8th edition, Page nos 1551, 1553, 1670, 1671
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page nos 127, 328, 1187
Online Resource, https://www.uptodate.com/contents/carbon-dioxide-monitoring-capnography
Miller's Anesthesia 9th edition, Page: 1392-1398, 1308, 1309,

Q80. Anesthesia NEET 2021 ID: MF1360

What is the maneuver being performed in the given image?



A ✓ Head tilt, chin lift

B Jaw thrust

C Head extension

D In line manual stabilisation

Correct Answer: Head tilt, chin lift

The maneuver being performed in the given image is that of **head-tilt chin-lift.** It is an important technique in emergency airway management to open up the airway.

Basic life support involves early recognition and management of emergencies such as heart attack, stroke, and foreign body airway obstruction. The sequence of BLS is **circulation** (chest compression), **airway**, and then **breathing.**

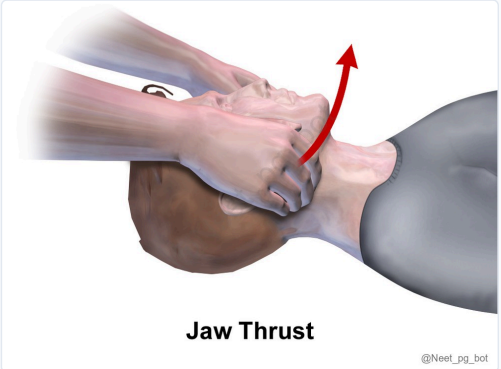
The airway is usually obstructed by posterior displacement of the tongue or epiglottis. One hand (palm) is placed on the patient's forehead applying pressure to tilt the head back. Simultaneously, the chin is lifted with the forefinger and index finger of the opposite hand. This causes cervical flexion with an atlantooccipital extension (sniffing position). This lifts the tongue from the back of the throat.

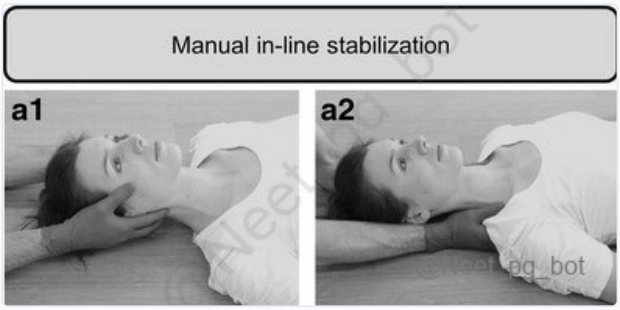
Head-tilt chin-lift should not be performed in a patient with suspected cervical injury.

Other options:

Option B: **Jaw thrust**- Both hands of the clinician are placed on either side of the patient's head and the angles of the jaw are grasped and lifted. It can be performed in a patient with cervical injury but caution must be followed to not hyperextend the neck.

Option D: **Manual inline stabilization** is a maneuver that limits external head and neck movement in patients with cervical spine injuries. Fingers and palms of both hands are used to stabilize the patient's occiput and mastoid processes to ensure successful airway intervention.





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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page nos 1232,1235
Online Resource, https://reference.medscape.com/features/slideshow/airway-management#page=3
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3982371/
Miller's Anesthesia 8th edition, Page 3184, 3185
Online Resource, https://www.msdmanuals.com/en-in/professional/critical-care-medicine/how-to-do-basic-airway-procedures/how-to-do-head-tilt-chin-lift-and-jaw-thrust-maneuvers

Q81. Anesthesia NEET 2021 ID: MF1349

A patient presented with headache, vomiting, and fever. Meningitis was suspected. While performing a lumbar puncture, which is the last structure you will pierce just before entering the lumbar cistern?

A Ligamentum flavum

B Dura mater

C ✓ Arachnoid membrane

D Pia mater

Correct Answer: Arachnoid membrane

The **arachnoid membrane** is the last structure you will pierce while performing a lumbar puncture, just before entering the lumbar cistern.

Lumbar puncture is a diagnostic procedure done to obtain a sample of **cerebrospinal fluid** from the **subarachnoid space.** During the procedure, the following structures are pierced to reach the subarachnoid space, in the given order:

- Skin
- Subcutaneous tissue
- Supraspinous ligament
- Interspinous ligament
- Ligamentum flavum
- Dura mater
- Arachnoid membrane

Indications:

- Diagnosis of suspected **meningitis** and **subarachnoid hemorrhage**
- **Spinal anesthesia** and intrathecal administration of antibiotics/chemotherapy
- Therapeutic relief of pseudotumor cerebri, less frequently

Contraindications for lumbar puncture include:

- Thrombocytopenia or other bleeding diatheses (including anticoagulant therapy)
- Raised intracranial pressure
- Spinal epidural abscess
- Infection over the site

Site of the procedure:

In adults, the spinal cord ends at the level of the lower border of the L1 vertebra. Lumbar puncture is usually performed at **L3-L4 space** (below L1 vertebra) to avoid potential needle trauma to the spinal cord.

Technique:

The patient is usually placed in the **lateral recumbent position**. An alternative position is the sitting position, useful in obese patients to determine the midline. L3-L4 space is located by palpating the right and left posterior superior iliac crests and moving the fingers medially toward the spine at the same level.

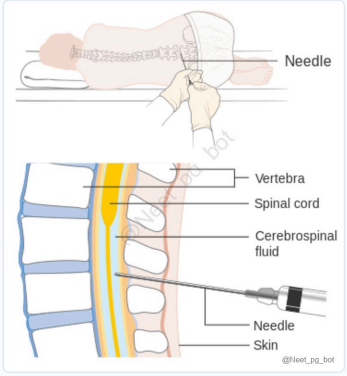
A local anesthetic like 1% lidocaine is administered. A 22-gauge needle is advanced through the skin. In the lateral recumbent position, the bevel should face up, and in the sitting position, it should face to one side or the other, to avoid injury to the dural fibers. There is a loss of resistance as the needle enters the epidural space. Dura and then arachnoid are next pierced, to release CSF.

Complications following lumbar puncture:

Post-dural puncture headache (PDPH) is a common post-procedure symptom. It occurs due to loss of CSF through the dura which causes traction on the pain-sensitive intracranial structures. The headache in PDPH is typically **frontal** or **occipital.** The smaller the needle used, the lesser the risk of PDPH.

Other complications include infections, bleeding, cerebral herniation, and radicular pain.

The image below shows a patient undergoing lumbar puncture in the lateral recumbent position:



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References: Gray's Anatomy: The Anatomical Basis of Clinical Practice, 41st Edition, Page no 773.e1
Miller's Anesthesia 8th edition, Page no 1685,1700, 1712
Online Resource, https://www.uptodate.com/contents/lumbar-puncture-technique-indications-contraindications-and-complications-in-adults?search=lumbar-puncture-technique-indications-&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
Online Resource, https://emedicine.medscape.com/article/80773-overview#a1

Q82. Anesthesia NEET 2021 ID: MF1358

A young male patient was given regional block with 0.25% bupivacaine for a minor surgical procedure. Shortly after the injection, the patient suddenly became unresponsive and the pulse was not recordable. Which of the following would be the best course of management for this patient?

A ✓ CPR with 20% intralipid

B CPR with sodium bicarbonate

C CPR with dobutamine

D CPR with calcium

Correct Answer: CPR with 20% intralipid

The clinical picture of an unresponsive patient with an absent pulse following bupivacaine administration is most likely due to **bupivacaine-induced cardiotoxicity** resulting from accidental intravascular injection of local anesthetic. The best course of management in this scenario would be administering **CPR (cardiopulmonary resuscitation)**, along with a **20% intralipid injection.**

Bupivacaine is a long-acting local anesthetic that is the most **cardiotoxic** among the local anesthetics. It depresses the rapid phase of depolarization in Purkinje fibers and ventricular muscle due to a decreased availability of fast sodium channels in cardiac membranes. This reduced rate of depolarization **depresses myocardial contractility.** Hence, bupivacaine is a potent cardio depressant.

Cardio-vascular effects of bupivacaine-induced toxicity are as follows:

- Left ventricular depression
- Atrioventricular heart block
- Ventricular arrhythmias and ventricular fibrillation
- Sinus bradycardia can lead to cardiac arrest

Initial management includes immediate initiation of **cardiopulmonary resuscitation** (CPR) and **defibrillation** as per the **advanced cardiac life support** (ACLS) algorithm.

Following CPR, **intralipid therapy** is initiated. Its mechanism is explained by the **lipid sink theory** which states lipophilic drugs (bupivacaine) are absorbed into the lipid emulsion of the plasma, hence, removed from tissues affected by toxicity.

A rapid bolus of **intralipid 20% IV** (1.5 mL/kg) is administered without delay, followed by an infusion of 0.25 mL/kg/min. The infusion should be given continuously for at least 10 minutes after hemodynamic stability is achieved.

Local anesthesia systemic toxicity (LAST) can also affect the central nervous system (CNS), which is more susceptible to local anesthetic action as compared to the cardiovascular system. Dizziness, tinnitus, and disorientation are the initial symptoms. Shivering, muscular twitching, and tremors follow. **Tonic-clonic seizures** occur. Finally, respiratory depression can lead to respiratory arrest.

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References: Miller's Anesthesia 8th edition, Page 1047, 1048
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page 274
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6001420/

Q83. Anesthesia NEET 2018 ID: MC3996

What is the mechanism of action of the curare group of muscle relaxants?

A Persistently depolarizing at neuromuscular junction

B ✓ Competitively blocking the binding of ACh to its receptors

C Repetitive stimulation of ACh receptors on muscle endplate

D Inhibiting the calcium channels on presynaptic membrane

Correct Answer: Competitively blocking the binding of ACh to its receptors

The **curare** group of drugs includes **non-depolarizing** neuromuscular blocking agents that act by **competitively blocking** the **binding** of **acetylcholine** to its receptors.

This leads to a decrease in the effect of acetylcholine and thus, neuronal transmission to the muscle does not occur.

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References: Miller's Anesthesia 8th edition, Page nos 959, 960, and 965

Q84. Anesthesia NEET 2018 ID: MC3994

Which of the following nerves is popularly used in clinical anaesthesia to monitor neuromuscular blockade?

A ✓ Ulnar Nerve

B Median Nerve

C Radial nerve

D Mandibular nerve

Correct Answer: Ulnar Nerve

Ulnar nerve stimulation at wrist is popularly used to assess the **adequacy of neuromuscular blockade.**

Neuromuscular blockade is measured using a peripheral nerve stimulator.

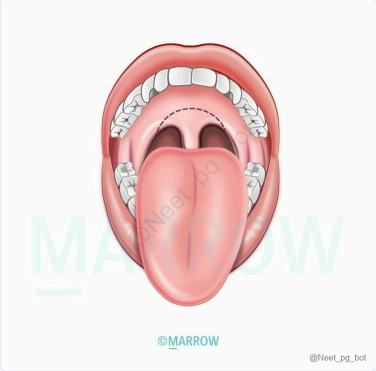
Nerve	Response
Ulnar nerve	Contraction of adductor pollicis
Facial nerve	Contraction of orbicularis oculi
Posterior tibial nerve	Contraction of flexor hallucis brevis

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References: Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5063182/
Miller's Anesthesia 8th edition, Page no 1610

Q85. Anesthesia NEET 2018 ID: MC3993

Identify the Mallampatti class of this patient.



A ✓ Class II

B Class I

C Class IV

D Class III

Correct Answer: Class II

The image shows Mallampatti class II airway.

The **modified Mallampatti classification** is used in the assessment of airway. It is a predictor of the difficulty of intubation or **difficult airway**.

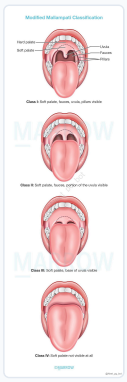
The assessment is performed with the patient sitting up straight, mouth open and tongue maximally protruded, **without saying "ahh."**

Modified Mallampati classification:

- Class 0: Ability to see any part of the epiglottis.
- Class I: Soft palate, fauces, uvula visible.
- Class II: Soft palate and fauces seen, tip of uvula not visible.
- Class III: Soft palate, base of uvula visible.
- Class IV: Soft palate not visible at all. Only hard palate visible.

Classes 3 and 4 are difficult intubation situations.

Initially, the Mallampatti scoring system (1985) had only 3 classes. This was modified by Samsoon and Young to include class 4. Later class 0 was also added.





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References: Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3045355/
Online Resource, https://emedicine.medscape.com/article/2172419-overview

Q86. Anesthesia NEET 2018 ID: MA2294

In which of the following devices is Murphy's eye seen?

A Nasopharyngeal airway

B Laryngoscope

C Bronchoscope

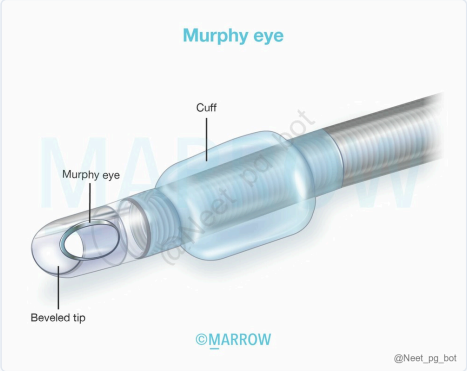
D ✓ Endotracheal tube

Correct Answer: Endotracheal tube

Murphy's eye is seen in the **endotracheal tube (ETT).**

Murphy's eye refers to the **distal opening** in the side wall of the ETT**.**

If the distal end of the lumen of the tube is blocked by secretions, Murphy's eye serves to provide **an alternate route for ventilation.**



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References: Miller's Anesthesia 8th edition, Page no. 1665, 2884
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page no. 321

Q87. Anesthesia NEET 2018 ID: MC3995

What is the most effective circuit for anaesthesia under spontaneous breathing?

A ✓ Mapleson A

B Mapleson B

C Mapleson C

D Mapleson D

Correct Answer: Mapleson A

The most effective circuit for anesthesia under **spontaneous** breathing is **Mapleson A** or **Magill's circuit**.

When the Mapleson A circuit is used for spontaneous ventilation, the fresh gas flow rate of the circuit is closest to the physiological minute ventilation of the patient. Hence it is preferred over other circuits.

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References: Miller's Anesthesia 8th edition, Page no 791
Miller's Anesthesia 9th edition, Page no. 611

Q88. Anesthesia NEET 2018 ID: MC4467

The best type of anesthesia for pre eclampsia is:

A Spinal anesthesia

B ✓ Epidural anesthesia

C General anesthesia

D Caudal block

Correct Answer: Epidural anesthesia

Continuous epidural anesthesia is the first choice for most patients with preeclampsia during labor, vaginal delivery, and cesarean section.

Pre-eclampsia patients have a risk of **severe airway edema**, which makes intubation difficult. Thus continuous epidural anesthesia is advantageous as it avoids the need for intubation. Besides, continuous epidural anesthesia can **improve uteroplacental perfusion** and also decrease catecholamine secretion.

But one major prerequisite for regional anesthesia in preeclamptic patients is normal platelet count and normal coagulation. Coagulopathy is a contraindication for regional anesthesia.

Minimal platelet count for regional anesthesia is 100,000/µL, but some cases can be taken up with a platelet count as low as 70,000/µL.

General anesthesia is indicated in patients with **coagulopathy** and in **severe pre-eclampsia** with complications.

Spinal anesthesia can be given in mild pre-eclampsia but hypotension associated with spinal anesthesia limits its use in moderate to severe pre-eclampsia.

Since the severity of pre-eclampsia or the type of delivery (vaginal or cesarean) is not mentioned, epidural anesthesia has an advantage over other techniques for above-mentioned reasons. Hence the best option here would be epidural anesthesia.

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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Pg 866
Morgan and Mikhail's Clinical Anesthesiology, 7th edition, page 1647

Q89. Anesthesia NEET 2018 ID: MD7321

What alternative drug would you use for cardiac arrest if epinephrine is not effective?

A Vasopressin

B Atropine

C ✓ Amiodarone

D Adenosine

Correct Answer: Amiodarone

Epinephrine and amiodarone are the two main drugs **approved by the AHA** for the resuscitation of a cardiac arrest patient.

The first dose of epinephrine 1 mg IV is administered after the 2nd defibrillator shock is delivered and it is continued every 3-5 minutes.

If this is not effective, 300 mg IV bolus dose of amiodarone can be given and another dose of 150 mg given if necessary

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References: Online Resource, American Heart Association Basic Cardiovascular Life Support Provider Manual- 2015 Guidelines Page no. 165, 175
Online Resource, https://www.acls.net/images/algo-arrest.pdf
Harrison's Principles of Internal Medicine 20th Edition, Page no: 2063
Online Resource, https://cpr.heart.org/en/resuscitation-science/cpr-and-ecc-guidelines/algorithms

Q90. Anesthesia NEET 2025 ID: MH0240

1. A 40-year-old female, on administration of a neuromuscular blocker, developed erythema, flushing of face, neck, trunk, and hypotension. Which of the following drugs is responsible?

A Pancuronium

B ✓ Atracurium

C Vecuronium

D Cis-atracurium

Correct Answer: Atracurium

The given clinical scenario is suggestive of histamine release. Atracurium is the most potent histamine releaser among the given options and was likely administered. Based on the chemical structure, non-depolarising muscle relaxants (NDMR) can be classified into two groups: * Steroidal compounds: Pancuronium, Vecuronium, Rocuronium * Benzylisoquinoline compounds: d-Tubocurarine, Atracurium, Cis-atracurium, and Mivacurium Erythema, flushing, and hypotension are manifestations of histamine release. Histamine release is most likely associated with benzylisoquinoline compounds, except cis-atracurium, as compared to steroidal compounds. Cis-atracurium is a stereoisomer of atracurium and has negligible histamine-releasing potential despite being a benzylisoquinoline compound. Management includes stopping or slowing the infusion of the NDMR and giving antihistamines. Hypotension can be prevented by giving antihistamines and non-steroidal anti-inflammatory drugs (NSAIDs). Pearl PM0247: Muscle relaxant with histamine release. Muscle relaxants causing histamine release leading to hypotension, tachycardia, flushing: Most likely: d-tubocurarine Likely: * Atracurium * Mivacurium * Succinyl choline Least likely: * Vecuronium * Cisatracurium * Rocuronium * Pancuronium

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References: Miller's Anesthesia 9th edition • Page no.799-800, 814-815

Q91. Anesthesia NEET 2025 ID: MH0241

2. Following a building collapse, a male patient was brought with a copious amount of debris in the mouth. The following procedure was done. Which of the following statements regarding the given procedure is true?



A It is used for effective ventilation upto 6 hours

B It is used to measure central venous pressure

C Work of breathing is more than bag and mask ventilation

D ✓ It should be followed by tracheostomy

Correct Answer: It should be followed by tracheostomy

The true statement regarding the given procedure is that it should be followed by tracheostomy. In this case, the presence of copious oral debris makes intubation and ventilation difficult, so a surgical airway is required. The image shows cricothyrotomy, a temporary emergency airway that must be followed by tracheostomy because of risks such as perichondritis, subglottic edema, and laryngeal stenosis. If intubation fails, options include a laryngeal mask airway, laryngeal tube airway, or surgical airway. Surgical airway is indicated in airway obstruction from glottic edema, laryngeal fracture, or severe oropharyngeal bleeding. Cricothyrotomy can be of two types: * Needle cricothyrotomy: A needle is introduced through the cricothyroid membrane to provide short-term ventilation. This may be combined with percutaneous transtracheal oxygenation at 15 L/min using a cannula and Y-connector. Intermittent insufflation (1 sec on, 4 sec off) provides oxygenation but inadequate CO2 clearance, so it is limited to 30-45 minutes. * Surgical cricothyrotomy: A transverse incision is made over the cricothyroid membrane, and a tube is inserted. It is preferred in emergencies over tracheostomy as it is faster, easier, and associated with less bleeding. Other options: Option A: Surgical cricothyrotomy can be used for effective ventilation up to 24 to 48 hours. Needle cricothyrotomy can be used only for 30 to 45 minutes. Option B: Cricothyrotomy is used for emergency airway management and not for the measurement of central venous pressure. Option C: In cricothyrotomy, the upper airway is bypassed, resulting in less resistance to airflow. Therefore, the work of breathing in cricothyrotomy is less, not more, than with bag-mask ventilation.

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References: P L Dhingra Diseases of Ear Nose and Throat and Head and neck surgery - 8th Edition • Page no. 374
Scott-Brown's Otorhinolaryngology and Head and Neck Surgery 8th Edition - 2018 • Volume 1 - Page no. 319-321, 357
Advanced Trauma Life Support ATLS Student Course Manual 2018 • Page no. 36

Q92. Anesthesia NEET 2025 ID: MH0237

3. A 50-year-old male with COVID developed ARDS and is on mechanical ventilation, with a PaO_2/FiO_2 ratio of 100. What is the most appropriate ventilator strategy for this patient?

A ✓ High PEEP and low tidal volume

B Low PEEP and low tidal volume

C High PEEP and high tidal volume

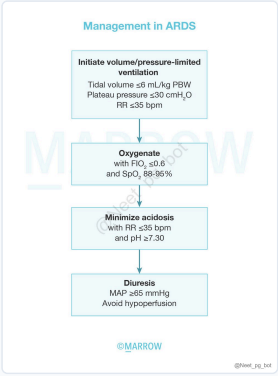
D Low PEEP and high tidal volume

Correct Answer: High PEEP and low tidal volume

The ventilator setting recommended for ARDS is high PEEP and low tidal volume. Acute respiratory distress syndrome (ARDS) is a clinical syndrome of severe dyspnea of rapid onset, occurring within 7 days of clinical insult, hypoxemia, bilateral pulmonary infiltrates, and may not demonstrate cardiomegaly, pleural effusions, leading to type 1 respiratory failure. Lung injury may be direct, as occurs in toxic inhalation, or indirect, as occurs in sepsis PaO2/FiO2 ratio is the ratio of arterial oxygen partial pressure (PaO2) to the fraction of inspired oxygen (FiO2). It is a key indicator of oxygenation. ARDS severity is assessed using the PaO2/FiO2 ratio at a minimum PEEP of 5 cm H2O as per the Berlin Criteria:

Severity	PaO_2/FiO_2 (mmHg)
Mild	200 - 300
Moderate	100 - 200
Severe	**≤100**

Recommended therapies in ARDS with clinical or supportive evidence: * Low tidal volume: To avoid barotrauma due to overdistention of the alveoli. * High PEEP (positive end-expiratory pressure): This prevents the collapse of alveoli at the end of expiration and improves oxygenation. * Prone position * Extracorporeal membrane oxygenation (ECMO)



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References: Harrison's Principles of Internal Medicine, 22nd Edition • Page no. 2299-2305
Miller's Anesthesia 9th edition • Page no. 2657

Q93. Anesthesia NEET 2025 ID: MH0239

4. General anesthetic sevoflurane and succinylcholine were administered to a patient undergoing abdominal surgery. Post operatively, patient developed persistent muscle contractions, hyperthermia and tachycardia. Which of the following drugs is life-saving in this situation?

A Diazepam

B ✓ Dantrolene

C Paracetamol

D Propanolol

Correct Answer: Dantrolene

The life-saving drug for the given condition is Dantrolene. The clinical picture of hyperthermia, tachycardia, and muscle rigidity following administration of sevoflurane and succinylcholine is characteristic of malignant hyperthermia, a genetic disorder caused by mutations in the RyR1 receptor of skeletal muscle. When exposed to triggering agents, the defective receptor leads to uncontrolled calcium release from the sarcoplasmic reticulum, resulting in persistent muscle contraction, increased heat production, acidosis, hyperkalemia, and elevated end-tidal CO2. Patients with positive family history and patients with congenital myopathies are at increased risk for the development of this condition. Drugs triggering malignant hyperthermia: * Succinylcholine * Halothane * Isoflurane * Sevoflurane Dantrolene is a direct-acting skeletal muscle relaxant that acts on the RyR1 (ryanodine receptor) calcium channels in the sarcoplasmic reticulum of skeletal muscles and inhibits calcium-induced calcium release. This results in the uncoupling of membrane depolarization and muscle contraction, thus inhibiting muscle contraction. Once the diagnosis is made, the causative inhalational agent should be stopped immediately. Supportive management includes rapid cooling, inhalation of 100% oxygen, control of acidosis, and treatment of hyperkalemia. Pearl PM0183: Malignant hyperthermia key features Pathophysiology: * Prolonged opening of functionally altered ryanodine receptors, leading to uncontrolled release of calcium from the sarcoplasmic reticulum resulting in sustained muscle contraction. Drugs known to trigger malignant hyperthermia: * Inhaled anesthetics: Halothane, Isoflurane, Sevoflurane, Desflurane, Ether, Methoxyflurane * Depolarising muscle relaxant: Succinylcholine Clinical signs: * Increased metabolism: Increased EtCO2, decreased SpO2, metabolic acidosis * Increased sympathetic activity: Tachycardia, hypertension, arrythmia * Muscle damage: Masseter spasm, generalised rigidity, hyperkalemia, rhabdomyolysis, myoglobinuria * Hyperthermia: Fever, sweating Treatment: * Discontinuation of all anesthetic agents * Hyperventilation with 100% O2 * Dantrolene: 2.5 mg/kg intravenously every 5 min until the episode is terminated (upper limit - 10 mg/kg). * Correction of acid-base, electrolyte abnormality * Cooling the patient

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References: Miller's Anesthesia 9th edition • Page no. 1114-1118
Goodman & Gillman's The Pharmacological Basis Of Therapeutics - 14th Edition • Page no. 243

Q94. Anesthesia INI-CET 2022 ID: MF4933

Arrange the following anesthetic agents on the basis of their potency.

Nitrous oxide

Halothane

Isoflurane

Methoxyflurane

A ✓ Methoxyflurane> Halothane> Isoflurane> Nitrous oxide

B Methoxyflurane> Isoflurane> Halothane> Nitrous oxide

C Methoxyflurane> Nitrous oxide> Halothane> Isoflurane

D Halothane> Isoflurane> Nitrous oxide> Methoxyflurane

Correct Answer: Methoxyflurane> Halothane> Isoflurane> Nitrous oxide

The **potency** of the inhaled anesthetics given above can be arranged as **methoxyflurane> halothane> isoflurane> nitrous oxide**. It is due to the change in the agents' **minimum alveolar concentration (MAC)**.

The **minimum alveolar concentration (MAC**) of an inhaled anesthetic is the **lowest concentration** required to prevent movement in response to a defined noxious stimulus in 50% of subjects. It is a measure of the **potency** of inhalational anesthetics. So, the lower the MAC value, the lower the concentration of anesthetic drug required, implying the drug is more potent.

MAC values (expressed as a '%' of 1 atmosphere) of some anesthetic agents:

- **Nitrous oxide** with a MAC value of 104 is the **least potent** inhaled anesthetic
- **Methoxyflurane** – MAC- 0.2 (**most potent**)
- **Desflurane**: 6.0
- Sevoflurane– 2.05
- Isoflurane – 1.28
- **Halothane** – 0.75

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References: Miller's Anesthesia 9th edition, Page no 511
Morgan and Mikhail's Clinical Anesthesiology, 6th edition, Page no 155

Q95. Anesthesia INI-CET 2022 ID: MF4921

Which of the following local anaesthetic causes hypertension?

A ✓ Cocaine

B Lignocaine

C Bupivacaine

D Prilocaine

Correct Answer: Cocaine

The local anesthetic that causes **hypertension** is **cocaine**. This is due to the inhibition of norepinephrine at the adrenergic terminals leading to increased **sympathetic stimulation**. This effect occurs at all concentrations of cocaine and causes **hypertension** and **ventricular ectopies**.

Treatment is with the help of benzodiazepines, α-adrenergic antagonists and amiodarone.

All local anesthetics are mainly cardiodepressants causing reduced contractility and vasodilation. However, at **low** concentrations, all **local anesthetics** cause **vasoconstriction** due to the inhibition of nitric oxide. Only cocaine produces vasoconstriction and hypertension at all concentrations.

Accidental intravascular injection of local anesthetic like bupivacaine results in **bupivacaine-induced cardiotoxicity.** Management is by resuscitation followed by **intralipid therapy**. A rapid bolus of intralipid 20% IV is administered without delay.

Other adverse effects of local anesthetics include neurologic toxicity, respiratory depression, and hypersensitivity reactions. Nervous system manifestations are circumoral numbness, tongue paresthesia, dizziness, tinnitus, and disorientation. Finally, **respiratory depression** can lead to respiratory arrest.

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References: Morgan and Mikhail's Clinical Anesthesiology, 6th edition, Page no 269, 271- 273
Miller's Anesthesia 9th edition, Page no 885

Q96. Anesthesia INI-CET 2022 ID: MF7321

Which of the following is required for the cannulation of a newborn?

1. 22G needle
2. 24G needle
3. 5-mL syringe
4. Adhesive tape
5. Normal saline

A 1,3,4,5

B ✓ 2,3,4,5

C 2,4

D 1,4

Correct Answer: 2,3,4,5

For cannulation in newborns, we would require a **24G needle, 5 mL syringe, adhesive tape, and normal saline.**

Also, for a peripherally inserted central catheter (PICC), we use a 24G needle for >1 kg neonate, and for <1 kg neonate, we use a 28G needle. We need to avoid using 1 mL syringes for flushing as it may lead to catheter rupture; preferably, use 5 or 10 mL syringes. **Flushing** of the cannulation is done using 0.9% sodium chloride or so-called **normal saline** (NS).

Dressing change: We change PICC line dressing only when visibly soiled, damp, or loosened. For making it easier and sterile, two persons should perform a dressing change.

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References: AIIMS Protocols in Neonatology 2019, 2nd Edition, page 506
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4446496/#:~:text=In%20this%20context%20of%20rinsing,order%20to%20clean%20the%20catheter.

Q97. Anesthesia INI-CET 2022 ID: MF7552

Identify the incorrect statement in rapid sequence intubation?

A Preoxygenation with 100% oxygen for 3 min is done.

B High-volume low-pressure cuff is used.

C ✓ It is the method of choice in cardiac arrest patient.

D Induction agent and muscle relaxant are given simultaneously.

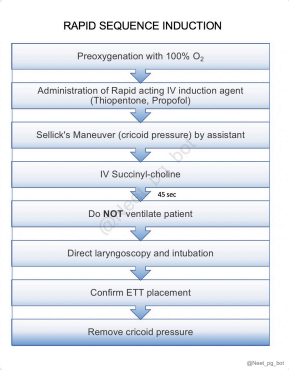
Correct Answer: It is the method of choice in cardiac arrest patient.

The **incorrect** statement regarding rapid sequence intubation (RSI) is that it is the **method of choice** in **cardiac arrest patients.**

RSI is a technique of anesthesia induction in patients who are at an **increased risk** of **gastric regurgitation** and **pulmonary aspiration** of gastric contents. Common cases include pregnancy, full stomach, intestinal obstruction, poorly controlled gastroesophageal reflux disease, morbid obesity, and diabetic gastroparesis. The goal is to intubate rapidly to minimize the time between the loss of consciousness and securing of the airway with a cuffed endotracheal tube.

It is achieved by **pre-oxygenating** the patient with **100% oxygen for 2-3 minutes**. An **inducing agent** is administered, which is **immediately followed** by a muscle relaxant succinylcholine or rocuronium. The trachea is intubated without any prior bag and mask ventilation using a **high-volume** and **low-pressure cuff** which adequately cuffs the airway and prevents aspiration.

Cardiac arrest is an **emergency condition** with importance given to start **chest compressions, defibrillation, or drugs** to restore perfusion. The airway is commonly blocked by posterior displacement of the tongue which can be managed by head tilt and chin lift or giving jaw thrust. In case of airway blockade or to prevent aspiration, the airway is maintained by emergency **tracheal tube intubation** and does not involve the procedure that is followed in RSI. Endotracheal tube placement is commonly done during **post-cardiac arrest** care.



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References: Miller's Anesthesia 9th edition, Page no. 1382
Online Resource, https://journals.sagepub.com/doi/pdf/10.1177/0310057X7700500215
Morgan and Mikhail's Clinical Anesthesiology, 6th edition, Page no.504
Online Resource, https://www.ahajournals.org/doi/10.1161/CIRCULATIONAHA.105.166557

Q98. Anesthesia NEET 2020 ID: MD0874

Which of the following investigations provides the most accurate prognostic information with respect to predicting risks of perioperative cardiac complications?

A Exercise ECG testing

B ✓ Dobutamine stress echocardiography

C Myocardial perfusion scintigraphy

D Coronary angiography

Correct Answer: Dobutamine stress echocardiography

Among the given investigations, **dobutamine stress echocardiography** provides the most accurate prognostic information with respect to predicting risks of perioperative cardiac complications.

Preoperative cardiac stress testing can help identify the presence and severity of preexisting CAD. The prognostic performance of these tests is better assessed based on positive and negative likelihood ratios. Among all the given options, dobutamine stress echocardiography has the maximum positive likelihood ratio and hence provides the most accurate prognostic information.

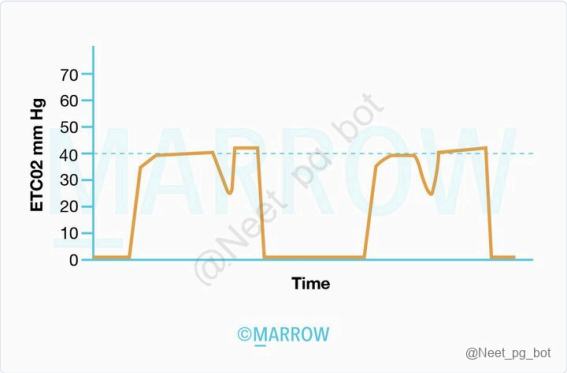
Even though coronary angiography is considered the gold standard, it may not absolutely risk-stratify patients because many perioperative cardiac events may be related to noncritical coronary lesions.

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References: Miller's Anesthesia 8th edition, Page No. 1095
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page No. 530
Miller's Anesthesia 9th edition, Page no. 1965

Q99. Anesthesia NEET 2020 ID: MD0678

A capnograph obtained from an intubated patient undergoing controlled ventilation is given below. Which of the following is a likely cause of the finding seen?



A Esophageal intubation

B Inspiration with cardiac oscillations

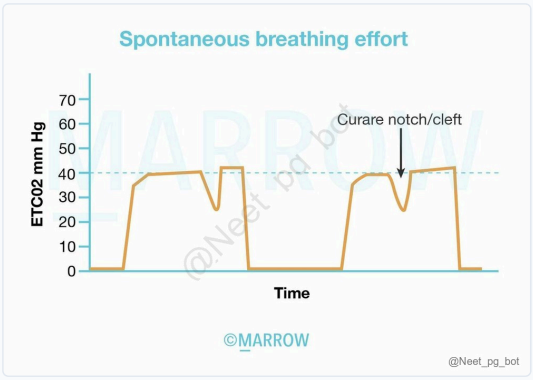
C Bronchospasm

D ✓ Spontaneous respiratory effort

Correct Answer: Spontaneous respiratory effort

The **curare notch** in the plateau phase of the capnograph indicates that the **patient is trying to breathe spontaneously**. The patient's inspiratory effort causes some fresh gas to be sucked in the capnometer, resulting in a lower carbon dioxide concentration and a notch.

Image below shows a capnograph with clefts during phase III which indicates **spontaneous breathing** efforts by the patient during controlled mechanical ventilation. It is also known as a **curare notch or cleft.**



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References: Miller's Anesthesia 8th edition, Page no 1554
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page nos 125, 127 and 128
Miller's Anesthesia 9th edition, Page no. 1311

Q100. Anesthesia NEET 2020 ID: MD0769

A 50-year-old patient with a history of hypertension and diabetes mellitus developed the following rhythm. His blood pressure was 90/40 mm Hg and he had a feeble pulse. What is the next step of management?



A IV adenosine

B IV diltiazem

C ✓ DC cardioversion

D Ibutilide

Correct Answer: DC cardioversion

The clinical scenario suggests **supraventricular tachycardia** with hemodynamic instability and the next step of management is **DC cardioversion** (50-100J).

If the patient was **hemodyamically stable,** initial treatment would be **vagal maneuvers** (Valsalva maneuver) in awake patients. This is because vagal maneuvers alone can terminate approximately 20% to 25% of reentry supraventricular tachycardias (SVTs).

If this is unsuccessful, then **iv adenosine 6 mg** is given. If necessary another dose of adenosine 12 mg may be administered after 1-2 minutes.

If the drug is administered via a central venous catheter, then these doses are reduced to 3 mg and 6 mg respectively.

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References: Miller's Anesthesia 8th edition, Page No. 1440
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page No. 483

Q101. Anesthesia NEET 2020 ID: MD0677

A patient is given a nicotinic receptor antagonist as a muscle relaxant. Which drug is given postoperatively to recover from muscle weakness?

A Physostigmine

B ✓ Neostigmine

C Carbachol

D Succinylcholine

Correct Answer: Neostigmine

Neostigmine is given postoperatively to **recover** from **muscle weakness.**

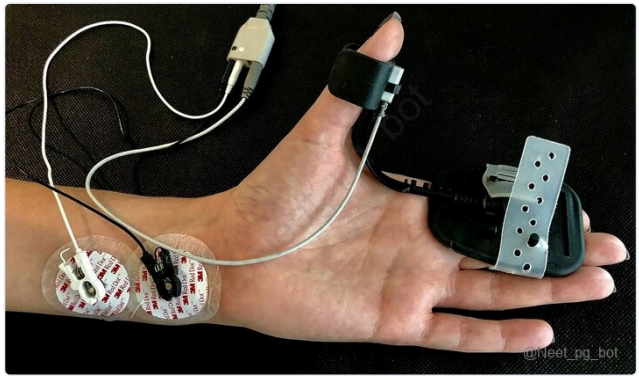
Non-depolarizing muscle relaxants **(NDMRs**) act mainly by the **competitive antagonism** of **nicotinic** (Nm) receptors. Neostigmine **inhibits acetylcholinesterase** and inhibits the degradation of acetylcholine. The increased accumulation of undegraded acetylcholine can effectively compete with NDMRs, and thereby displace them from the site and antagonize their effects.

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References: Miller's Anesthesia 8th edition, Page no. 425
Miller's Anesthesia 9th edition, Page no. 334

Q102. Anesthesia NEET 2022 ID: MF5335

The image given below shows neuromuscular monitoring of the patient after anesthesia. What is the most commonly used nerve for monitoring?



A ✓ Ulnar nerve

B Median nerve

C Radial nerve

D Metacarpal nerve

Correct Answer: Ulnar nerve

The following image shows neuromuscular monitoring and the most commonly used nerve for monitoring is the **ulnar nerve.**

The mechanosensor with electrodes is placed over the region of the **ulnar nerve.** This assesses the contraction of the **adductor pollicis** muscle to monitor neuromuscular activity.

Monitoring of neuromuscular blockade is required to **monitor the blockade intraoperatively**, ensure its complete reversal postoperatively, and **prevent residual blockade**. To monitor the neuromuscular function, a peripheral motor nerve is supramaximally stimulated and the response of the muscle is assessed.

The most commonly assessed nerve-muscle unit is the **ulnar nerve-adductor pollicis muscle**. This is because it is easily accessible, has the lowest risk of direct muscle stimulation, and the response can be evaluated visually, by tactile methods, or by objective methods. To stimulate the ulnar nerve, the electrodes are best applied to the **volar aspect** of the wrist.

Other nerve-muscle units that can be assessed include:

- Posterior tibial nerve- flexor hallucis brevis muscle
- Facial nerve- orbicularis oculi muscle
- Facial nerve- corrugator supercilii muscle

Many methods can be used for objective monitoring of neuromuscular function:

- Mechanomyography (MMG)
- Electromyography (EMG)
- Acceleration of the muscle response (AMG)
- Kinemyography (KMG)
- Cuff pressure modality (CPM)
- Compressomyography (CMG)
- Phonomyography (PMG)

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References: Miller's Anesthesia 9th edition, Page no 1356,1357, 1361

Q103. Anesthesia NEET 2022 ID: MF5309

Identify the mask in the image used for patients with COVID-19 infection.



A Venturi mask

B Hudson mask

C Nebuliser

D ✓ Non-rebreathing mask

Correct Answer: Non-rebreathing mask

The mask in the image is a **non-rebreathing mask** (NRBM). It can be used for patients with COVID-19 infection.

A NRBM has a **reservoir bag** attached to the mask in order to provide additional reservoir volume. In addition, there is a valve attached, which provides a unidirectional flow of gases. Flutter valves on the side ports prevent room air entrainment, while a valve incorporated between the mask and the reservoir bag prevents the entry of the exhaled gases.

NRBM can provide the **highest FiO_2** of 0.9 to 1.0. without intubation at the fresh gas flow of 12-15 L/min. They are used for preoxygenation before intubation and post-extubation in covid patients.

Other options:

Option A: The venturi mask is a **high-flow, fixed-performance device** that delivers a **specific oxygen concentration** to the patient. It is a color-coded device that connects to the oxygen tubing distally and has a nozzle connecting to a mask proximally. Oxygen flows through a central constriction inside the device. Room air enters through surrounding apertures which dilutes the oxygen to the required concentration. The venturi mask is based on Bernoulli's principle.

Option B: **Hudson mask** is a simple face mask that delivers oxygen without forming a tight seal against the patient's face. There are holes on either side of Hudson's mask. This provides air entrainment, meeting the peak inspiratory flow and venting out the expired gases.

Option C: A nebulizer is used to deliver drugs to the breathing system. It generates an aerosol, producing droplets of varying sizes. It is shown below.







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References: Morgan and Mikhail's Clinical Anesthesiology, 6th edition, Page no 1333-1334
Equipment in Anaesthesia and Critical Care, Page no 18, 20
Online Resource, https://www.bjoaonline.com/article.asp?issn=2549-2276;year=2020;volume=4;issue=5;spage=3;epage=7;aulast=Roy
Online Resource, https://openanesthesiajournal.com/VOLUME/11/PAGE/35/FULLTEXT/

Q104. Anesthesia NEET 2023 ID: MF8078

Identify the maneuver being performed in the given image.



A ✓ Head tilt, chin lift

B Jaw thrust

C Head extension

D In line manual stabilization

Correct Answer: Head tilt, chin lift

The maneuver performed in the given image is that of a **head-tilt chin-lift.** It is an important technique in emergency airway management to open up the airway.

Basic life support (BLS) involves early recognition and management of emergencies such as heart attack, stroke, and foreign body airway obstruction. The sequence of BLS is **circulation** (chest compression), **airway**, and **breathing.**

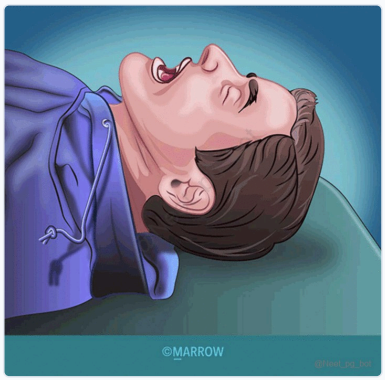
The airway is usually obstructed by posterior displacement of the tongue or epiglottis. One hand (palm) is placed on the patient's forehead, applying pressure to tilt the head back. Simultaneously, the chin is lifted with the forefinger and index finger of the opposite hand. This causes cervical flexion with an atlantooccipital extension (sniffing position). This lifts the tongue from the back of the throat.

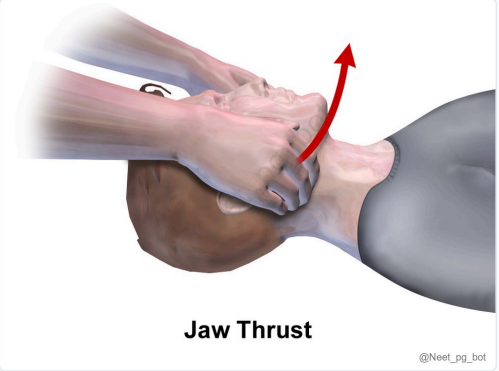
Head-tilt chin-lift should **not be performed in a patient with suspected cervical injury.**

Other options:

Option B: **Jaw thrust** - Both hands of the clinician are placed on either side of the patient's head, and the angles of the jaw are grasped and lifted. It can be **performed in a patient with cervical injury,** but caution must be exercised to not hyperextend the neck.

Option D: **Manual inline stabilization** is a maneuver that limits external head and neck movement in patients with cervical spine injuries. Fingers and palms of both hands are used to stabilize the patient's occiput and mastoid processes to ensure successful airway intervention.







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References: Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page nos 1232,1235
Online Resource, https://reference.medscape.com/features/slideshow/airway-management#page=3
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3982371/
Miller's Anesthesia 8th edition, Page 3184, 3185
Online Resource, https://www.msdmanuals.com/en-in/professional/critical-care-medicine/how-to-do-basic-airway-procedures/how-to-do-head-tilt-chin-lift-and-jaw-thrust-maneuvers
Miller's Anesthesia 9th edition, Page no 2157

Q105. Anesthesia NEET 2023 ID: MF8120

Which of the following is the drug of choice for preoperative antibiotic prophylaxis in a patient undergoing cardiac surgery?

A Penicillin G

B Erythromycin

C Azithromycin

D ✓ Cefazolin

Correct Answer: Cefazolin

First- and second-generation cephalosporins (cefazolin) are the drugs of choice for preoperative prophylaxis for cardiac surgery. **Clindamycin** is reserved for cases of **allergy to beta-lactams** and **vancomycin** is recommended for **methicillin-resistant *Staphylococcus aureus* (MRSA).**

Preoperative antibiotic prophylaxis is essential in **reducing the risk of surgical site infections**, particularly in cardiac surgery, where the consequences of infection can be severe. The choice of antibiotic depends on the patient's medical history, allergies, and the most common pathogens involved in surgical site infections for the specific surgery. In addition to cefazolin, other cephalosporins, such as cefuroxime, can also be used for prophylaxis in cardiac surgery. The **timing of antibiotic administration is crucial**, with the **first dose** typically given **within 60 minutes before the incision.**

Other Options:

Option A: Penicillin G is not the drug of choice for preoperative antibiotic prophylaxis in cardiac surgery. It is mainly used to treat infections caused by susceptible gram-positive bacteria, such as *Streptococcus pyogenes*, *Streptococcus pneumoniae*, and some strains of *Staphylococcus aureus*.

Option B: Erythromycin is a macrolide antibiotic and is not the drug of choice for preoperative prophylaxis in cardiac surgery. It is typically used to treat respiratory tract infections, skin infections, and other infections caused by susceptible gram-positive bacteria.

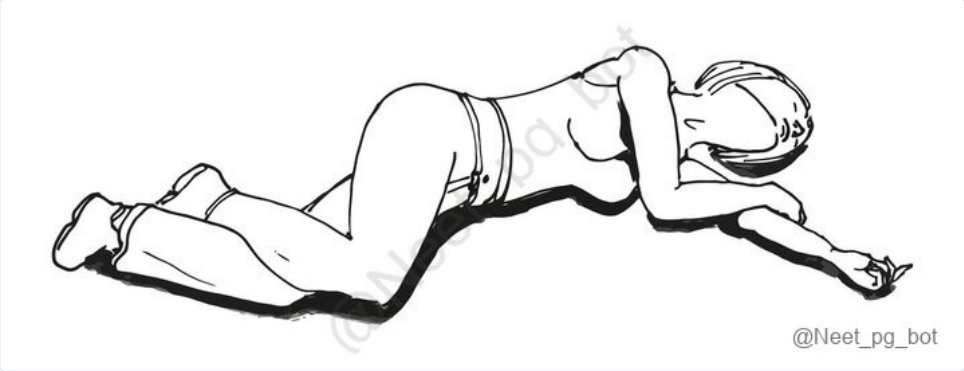
Option C: Azithromycin, another macrolide antibiotic, is also not the drug of choice for preoperative prophylaxis in cardiac surgery. It is commonly used to treat respiratory tract infections, sexually transmitted infections, and other infections caused by susceptible gram-positive and gram-negative bacteria.

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References: Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4782846/#:~:text=The%20main%20criteria%20of%20prophylaxis,are%20the%20antibiotics%20of%20choice. Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK442032/#:~:text=The%20three%20antibiotics%20used%20in,who%20should%20receive%203%20g.

Q106. Anesthesia NEET 2023 ID: MF8128

During resuscitation, when is the given position indicated?



A Unconsciousness with pulse and breathing absent

B Unconsciousness with pulse present and breathing absent

C ✓ Unconsciousness with pulse and breathing present

D Unconsciousness with pulse absent and breathing present

Correct Answer: Unconsciousness with pulse and breathing present

The position illustrated in the picture is known as the **recovery position,** and it is used in the management of patients who are **unconscious** but with **pulse present** and **breathing present.**

The recovery position can help prevent complications such as aspiration and airway obstruction in unconscious individuals. Additionally, it is crucial to monitor the patient's airway, breathing, and circulation continuously while they are in the recovery position and seek immediate medical assistance. An unresponsive patient who is breathing spontaneously in the supine position is at risk of airway obstruction from their tongue, mucus, vomit, etc. A modified lateral position is used and the following 6 principles are considered while choosing which variation of recovery position to use,

1. The victim should be in as near a true **lateral position** as possible, with the head dependent to allow free drainage of fluid.
2. The position should be stable.
3. **Avoid any pressure** on the chest that impairs breathing.
4. It should be possible to turn the victim on his or her side and to return to the back easily and safely, with concern for a possible cervical spine injury.
5. Good observation of and access to the airway should be possible.
6. The position itself should not cause injury to the victim.

Other options

Option A: Unconsciousness with pulse and breathing absent - In this situation, the patient is in cardiac arrest and requires immediate cardiopulmonary resuscitation (CPR), not the recovery position. CPR involves chest compressions and rescue breaths to support circulation and respiration.

Option B: Unconsciousness with pulse present and breathing absent - In this case, the patient is experiencing respiratory arrest. Rescue breathing or bag-valve-mask ventilation should be initiated to support the patient's breathing. The recovery position is not indicated in this scenario.

Option D: Unconsciousness with pulse absent and breathing present - This option is not physiologically plausible, as the absence of a pulse implies a lack of circulation, and breathing cannot be present without circulation.

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References: Online Resource, https://www.ahajournals.org/doi/full/10.1161/circ.102.suppl_1.I-22#d3e1950

Q107. Anesthesia AIIMS 2018 ID: MC5207

A patient is undergoing modified radical neck dissection (MRND) for laryngeal malignancy. While ligating the internal jugular vein, his etCO2 dropped from 40 mmHg to 14 mmHg, and the patient developed hypotension along with cardiac arrhythmia. Which of the following is the most likely cause?

A Sympathetic overactivity

B Vagal stimulation

C ✓ Venous air embolism

D Carotid body stimulation

Correct Answer: Venous air embolism

A drastic **fall in $etCO_2$ with hypotension during neck surgery** is most likely due to **venous air embolism (VAE).**

In modified radical neck dissection (MRND), a head-up position is given to provide a bloodless surgical field. VAE into severed vessels is likely to occur when the surgical site is above the right atrium and venous pressure at the incision site is below the atmospheric pressure. Pulmonary artery pressure will rise and $etCO_2$ will fall after a venous air embolism.

In sympathetic overactivity and carotid body stimulation, blood pressure will rise.

In vagal stimulation, a drastic drop in $etCO_2$ is unlikely.

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References: Miller's Anesthesia 8th edition, Page no 2169
Miller's Anesthesia 9th edition, Page no. 1878

Q108. Anesthesia AIIMS 2018 ID: MC5206

At what level does the spinal cord end in adults?

A L3-L4

B ✓ L1-L2

C T12-L1

D L2-L3

Correct Answer: L1-L2

The distal termination of the spinal cord is the **lower border of L1 in adults.**

Because the spinal cord ends at L1, lower nerve roots course some distance before exiting the intervertebral foramina. These lower spinal nerves form the cauda equina.

Note:

In children, the spinal cord ends at L3 and moves up with age.

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References: Miller's Anesthesia 8th edition, Page no 1686
Morgan and Mikhail's Clinical Anesthesiology, 5th Edition, Page nos 944 and 945

Q109. Anesthesia AIIMS 2018 ID: MA5004

Which of the following is not a contraindication to the use of morphine?

A Bronchial asthma

B Renal failure

C Emphysema

D ✓ Acute myocardial infarction

Correct Answer: Acute myocardial infarction

Acute myocardial infarction is **not** a contraindication for morphine use.

Morphine is the **opioid of choice** for **myocardial infarction** as it relieves anxiety, chest pain, and myocardial oxygen demand.

Contraindications to the use of Morphine:

- **Bronchial asthma:** Histamine release can cause bronchoconstriction and precipitate attacks of asthma.
- **Renal failure:** Morphine-6-glucuronide (metabolite) can accumulate and lead to prolonged respiratory depression.
- **Emphysema:** Respiratory depression and decreased response to hypoxia can cause hypercapnia in patients with COPD
- **Head injury:** Hypercapnia due to respiratory depression can increase ICP. Miosis, vomiting, and effect on wakefulness make interpreting the signs of head injury difficult.
- **Biliary colic:** Constriction of sphincter of Oddi can worsen biliary spasm
- **Hypotension** and shock

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References: Miller's Anesthesia 8th edition, Page no. 878, 891
Goodman & Gillman's The Pharmacological Basis Of Therapeutics - 13th Edition, Page no. 366-368

Q110. Anesthesia AIIMS 2018 ID: MA2738

Cardiac output can be determined using the following except?

A Thermal dilution technique

B ✓ Capnography

C Esophageal doppler

D Echocardiography

Correct Answer: Capnography

Capnography is a measurement of **end-tidal CO_2**, which is used to confirm tracheal intubation and assess ventilation in the perioperative period.

Cardiac output measurements technique:

1. Thermodilution
2. Dye dilution, using indocyanine green
3. Lithium dilution
4. Pulse contour devices
5. Esophageal doppler
6. Thoracic bioimpedance
7. Echocardiography

Note: Thermal dilution method uses a non-toxic liquid of temperature different from the blood. This fluid is injected into a vein and is detected on the arterial side using a catheter with a temperature sensor. The rate of flow of this liquid is used to calculate the cardiac output.

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References: Miller's Anesthesia 8th edition, Page nos 1387-90
Online Resource, https://pubmed.ncbi.nlm.nih.gov/20185659/
Online Resource, https://www.sciencedirect.com/topics/medicine-and-dentistry/thermodilution